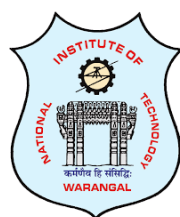


Lecture Notes

Dynamical Systems

(MA17029, MA1516)

PG Notes



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Preface

These lecture notes have been prepared for the course **Nonlinear Dynamics**. The notes are intended to supplement classroom lectures and provide students with a concise mathematical treatment of nonlinear dynamical systems, bifurcation theory, chaos, Hamiltonian systems, Lyapunov exponents, and related topics.

The material is compiled primarily for educational purposes. Students are encouraged to consult the prescribed textbooks and research articles for additional details and proofs.

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Course Information

Dynamical Systems

MA17029, MA1516

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Course Outcomes

CO-1	Identify Autonomous and Nonautonomous Systems.
CO-2	Understand Limit Cycle Motion and Periodic Attractor.
CO-3	Differentiate Dissipative and Conservative Systems.
CO-4	Understand Different Types of Bifurcations.
CO-5	Apply Poincaré-Bendixson Theory.

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Learning Resources

Text Books

1. M. Lakshmanan and S. Rajasekar, Nonlinear Dynamics: Integrability, Chaos and Patterns, Springer, First Edition, 2010.
2. George F. Simmons, Differential Equations with Applications and Historical Notes, McGraw-Hill, Second Edition, 2003.

Reference Books

1. Morris W. Hirsch, Stephen Smale and Robert L. Devaney, Differential Equations, Dynamical Systems, and an Introduction to Chaos, Elsevier Academic Press, 2004.
2. Lawrence Perko, Differential Equations and Dynamical Systems, Third Edition, Springer, 2010.

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1. Introduction

Nature is inherently dynamic. Every physical, biological, chemical and engineering system undergoes continuous evolution with time. The motion of planets, oscillations of a pendulum, propagation of electrical signals, chemical reactions, population growth, and the spread of infectious diseases are all examples of systems whose states change continuously with time. Understanding and predicting such changes form one of the principal goals of mathematical modelling.

The mathematical study of systems whose states evolve with time is known as **Dynamical Systems Theory**. It provides a unified framework for describing the evolution of a wide variety of natural and artificial systems. Depending on the application, the governing equations may be ordinary differential equations, partial differential equations, difference equations, delay differential equations, stochastic differential equations, or integro-differential equations.

Historically, the foundations of dynamical systems were laid by **Isaac Newton** through his laws of motion, which explained how the motion of an object is determined by the forces acting upon it. During the nineteenth century, the works of **Joseph-Louis Lagrange** and **William Rowan Hamilton** provided elegant reformulations of mechanics that remain fundamental in modern mathematics and theoretical physics.

Although many mathematical models appear similar, their qualitative behaviour depends strongly on whether the governing equations are *linear* or *nonlinear*. Linear systems possess several remarkable mathematical properties that make them relatively easy to analyze. Nonlinear systems, however, often exhibit complicated behaviours such as multiple equilibrium points, limit cycles, bifurcations and chaos. Consequently, nonlinear dynamical systems have become one of the most active areas of modern applied mathematics.

This course is devoted to understanding the mathematical theory underlying nonlinear dynamical systems.

1.1 What is a Dynamical System?

One of the fundamental questions in science is

“How does a system evolve with time?”

A mathematical description of this evolution is called a **dynamical system**.

Definition 1. A **dynamical system** is a mathematical model that describes the evolution of one or more state variables with respect to time according to a prescribed law.

If the state vector is denoted by

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix},$$

then a general continuous-time dynamical system can be written as

$$\dot{x} = f(x, t),$$

where

- $x(t) \in \mathbb{R}^n$ is called the **state vector**,
- t denotes time,
- $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ describes the governing law.

The function f determines how the state evolves with time.

Examples of Dynamical Systems: Dynamical systems arise naturally in almost every branch of science and engineering.

System	State Variables
Simple Pendulum	Angular displacement and angular velocity
Planetary Motion	Position and velocity
Electrical Circuit	Voltage and current
Population Model	Population size
Neural Network	Activations of neurons
Robot Manipulator	Joint positions and velocities
Financial Market	Asset prices
Weather Model	Temperature, pressure and humidity

Although these systems appear unrelated, they can all be represented mathematically using differential equations (or difference equations).

1.2 Continuous and Discrete Dynamical Systems

Dynamical systems are broadly classified into two categories.

Continuous-Time Systems A continuous-time system evolves continuously with respect to time and is generally described by an ordinary differential equation

$$\dot{x} = f(x, t).$$

Examples include

- motion of planets,
- electrical circuits,
- mechanical vibrations,
- fluid flow.

Discrete-Time Systems A discrete-time system evolves only at discrete instants

$$x_{k+1} = F(x_k).$$

Such systems naturally arise in

- digital control,

- numerical algorithms,
- economic models,
- population models with seasonal breeding.

Throughout these notes we shall mainly consider continuous-time systems but also discuss some concepts in discrete systems.

State Variables The smallest collection of variables that completely describes the present condition of a system is called the **state**.

Definition 2. *A set of variables is called a **state vector** if knowledge of these variables at an initial time together with the governing equations uniquely determines the future evolution of the system.*

Example For a simple pendulum the angular positions, $\theta(t)$ alone is insufficient because two pendulums may have the same angular position but different angular velocities. Hence the state vector is

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}.$$

Thus the pendulum is a two-dimensional dynamical system.

Why Nonlinear Dynamical Systems? Most physical systems encountered in practice are inherently nonlinear. Linear models are usually obtained by making simplifying assumptions valid only near an equilibrium point. While these approximations are mathematically convenient, they often fail to describe the complete behavior of the underlying physical system. Examples include large-angle pendulum motion, spacecraft attitude dynamics, fluid turbulence, ecological systems, climate dynamics, neural activity, robotics, and power systems. Nonlinear systems can exhibit remarkable phenomena that never occur in linear systems, including multiple equilibrium points, bifurcations, chaos, and sensitive dependence on initial conditions. The study of these behaviors forms the subject of nonlinear dynamics.

2. Qualitative Features of Dynamical Systems

For many **nonlinear dynamical systems**, obtaining an exact analytical solution is either extremely difficult or impossible. Consequently, instead of searching for explicit solutions, mathematicians and scientists often investigate the **qualitative behaviour** of solutions. Its aim is to answer questions such as

- Does the system eventually settle down to an equilibrium?
- Is the equilibrium stable or unstable?
- Does the system exhibit periodic oscillations?
- Can the system display complicated or chaotic motion?

Rather than computing the exact trajectory, qualitative analysis studies the overall behavior of solutions.

Why Qualitative Analysis? Most nonlinear differential equations cannot be solved exactly. However, valuable information can still be obtained by examining certain special solutions and their neighborhood. The simplest objects to study are Equilibrium (fixed) points, Periodic solutions, Limit cycles, and More complicated invariant sets such as attractors. These objects determine the long-term behaviour of many dynamical systems.

2.1 Periodic Motion

Not every dynamical system approaches an equilibrium. Many systems instead exhibit repeating motions. A solution satisfying

$$x(t + T) = x(t)$$

for some constant $T > 0$ is called a **periodic solution**.

Examples

- Oscillation of a pendulum.
- Planetary motion around the Sun.
- Vibrations of springs.
- Electrical oscillators.

Periodic solutions play an important role because they describe sustained oscillatory behavior.

Beyond Periodic Motion Some nonlinear systems exhibit behavior much more complicated than simple equilibria or periodic oscillations. Such systems may exhibit quasiperiodic, chaotic, or strange attractors. Although these motions are difficult to predict exactly, their overall geometric structure can still be studied.

2.2 Orders of Differential Equations

Physical systems may be described by differential equations of different orders.

Second-Order Equations Many mechanical systems obey Newton's second law and are naturally described by second-order differential equations.

Example:

$$m\ddot{x} + c\dot{x} + kx = 0.$$

Higher-Order Equations Some engineering systems contain third-order or higher-order equations, especially in control theory and beam dynamics.

First-Order Systems Many models in biology, chemistry, economics, and population dynamics are naturally expressed as systems of first-order differential equations.

Example:

$$\begin{aligned}\dot{x} &= f(x, y), \\ \dot{y} &= g(x, y).\end{aligned}$$

2.3 Autonomous and Nonautonomous Systems

Dynamical systems are broadly classified into two categories.

2.3.1 Autonomous Systems

An autonomous system has no explicit dependence on time. Its general form is

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}).$$

The governing equations remain unchanged with time.

Examples

- Exponential oscillator

$$\ddot{x} + e^x = 0.$$

- Lorenz equations

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz.\end{aligned}$$

2.3.2 Nonautonomous Systems

A nonautonomous system explicitly depends on time. Its general form is

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t).$$

The external forcing or time-dependent parameters make the dynamics change with time.

Examples

- Forced harmonic oscillator

$$\ddot{x} + a\dot{x} + \omega_0^2 x = f \cos(\omega t).$$

- Forced Van der Pol oscillator

$$\ddot{x} - b(1 - x^2)\dot{x} + \omega_0^2 x = f \cos(\omega t).$$

- Forced pendulum

$$\ddot{x} + a\dot{x} + \sin x = f \sin(\omega t).$$

Difference Between Autonomous and Nonautonomous Systems

Autonomous	Nonautonomous
No explicit time dependence	Explicit time dependence
Form: $\dot{x} = f(x)$	Form: $\dot{x} = f(x, t)$
System parameters remain constant	External forcing present
Phase portrait remains unchanged	Phase portrait changes with time
Examples: Lorenz system	Examples: Forced pendulum

2.4 Dynamical Systems as Coupled First-Order Differential Equations

Most dynamical systems arising in science and engineering are described by ordinary differential equations (ODEs). These equations may be linear or nonlinear, autonomous or nonautonomous, and of first or higher order. Fortunately, every finite-dimensional dynamical system can be represented as a system of coupled first-order differential equations. This representation is known as the **state-space representation** and forms the basis of modern dynamical systems theory, control theory, and numerical simulation.

Why Convert to First-Order Systems? The conversion of higher-order differential equations into first-order systems offers several advantages like that it provides a unified mathematical framework for studying all dynamical systems, the theory of phase space is naturally developed using first-order equations, stability, equilibrium points, and long-term behavior are easier to analyze and most numerical algorithms are designed for first-order systems.

Procedure for conversion of higher-order equations The general procedure is straightforward. Consider an n^{th} -order differential equation

$$x^{(n)} = f(x, \dot{x}, \ddot{x}, \dots, x^{(n-1)}, t).$$

Introduce new variables

$$x_1 = x, \quad x_2 = \dot{x}, \quad x_3 = \ddot{x}, \quad \dots, \quad x_n = x^{(n-1)}.$$

Then

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= x_3, \\ &\vdots \\ \dot{x}_{n-1} &= x_n, \\ \dot{x}_n &= f(x_1, x_2, \dots, x_n, t).\end{aligned}$$

Thus every higher-order equation becomes a system of first-order equations.

Example: Linear Harmonic Oscillator The equation of motion is

$$\ddot{x} + \omega_0^2 x = 0.$$

Introduce

$$y = \dot{x}.$$

The equivalent first-order system is

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -\omega_0^2 x.\end{aligned}$$

This system has two state variables,

$$X = (x, y)^T.$$

Example: Duffing Oscillator The Duffing oscillator is governed by

$$\ddot{x} + a\dot{x} + \omega_0^2 x + \beta x^3 = f \cos(\omega t).$$

Let

$$y = \dot{x}.$$

Then

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -ay - \omega_0^2 x - \beta x^3 + f \cos(\omega t).\end{aligned}$$

Since time appears explicitly, this is a **nonautonomous system**. To convert it into an autonomous system, introduce

$$z = \omega t.$$

Since

$$\dot{z} = \omega,$$

the autonomous system becomes

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -ay - \omega_0^2 x - \beta x^3 + f \cos z, \\ \dot{z} &= \omega.\end{aligned}$$

Example: Rigid Body Equations The rotational motion of a rigid body is described by Euler's equations

$$I_1 \dot{\omega}_1 - (I_2 - I_3) \omega_2 \omega_3 = N_1,$$

$$I_2 \dot{\omega}_2 - (I_3 - I_1) \omega_3 \omega_1 = N_2,$$

$$I_3 \dot{\omega}_3 - (I_1 - I_2) \omega_1 \omega_2 = N_3.$$

These equations are already first-order differential equations in the angular velocities

$$(\omega_1, \omega_2, \omega_3).$$

Hence no further conversion is required.

Example: Lotka-Volterra Model The predator-prey model is

$$\dot{x} = ax - xy,$$

$$\dot{y} = xy - by.$$

These equations are naturally expressed as a coupled first-order system. Here

- $x(t)$ denotes the prey population,
- $y(t)$ denotes the predator population.

General State-Space Representation Suppose a dynamical system contains n state variables

$$X = (x_1, x_2, \dots, x_n)^T.$$

Its evolution is represented by

$$\dot{X} = F(X),$$

where

$$F(X) = \begin{bmatrix} F_1(X) \\ F_2(X) \\ \vdots \\ F_n(X) \end{bmatrix}.$$

Equivalently,

$$\dot{x}_1 = F_1(x_1, x_2, \dots, x_n),$$

$$\dot{x}_2 = F_2(x_1, x_2, \dots, x_n),$$

$$\vdots$$

$$\dot{x}_n = F_n(x_1, x_2, \dots, x_n).$$

The vector X is called the **state vector**, and the space containing all possible states is called the **state space** or **phase space**.

State Variables The choice of state variables depends on the physical system.

Examples Position and velocity, Position and momentum, Angular velocity, Energy, Population densities, and Chemical concentrations. Once the state variables are selected, the complete future behavior of the system is determined by the current state.

2.5 Equilibrium (Fixed) Points

An equilibrium point is a state at which the system does not evolve with time. For

$$\dot{X} = F(X),$$

an equilibrium point satisfies

$$F(X^*) = 0.$$

Equivalently,

$$\dot{X} = 0.$$

If the system starts at an equilibrium point, it remains there forever. Therefore, equilibrium points are also called stationary points, fixed points, or steady states.

Procedure for Finding Equilibrium Points To determine the equilibrium points,

1. Write the system in first-order form.
2. Set every derivative equal to zero.
3. Solve the resulting algebraic equations simultaneously.

Example Consider the dynamical system

$$\dot{x} = x(1 - x).$$

The equilibrium points are obtained by solving

$$f(x_e) = x_e(1 - x_e) = 0.$$

Hence,

$$x_e = 0 \quad \text{or} \quad x_e = 1.$$

Therefore, the system has two equilibrium points:

$$x_e = 0, \quad x_e = 1.$$

For example, if the initial condition is $x(0) = 1$, then

$$\dot{x} = 1(1 - 1) = 0.$$

Thus, the state remains at $x(t) = 1$ for all $t \geq 0$. Similarly, if $x(0) = 0$, the state remains at $x(t) = 0$ for all $t \geq 0$.

If the system starts exactly at an equilibrium point, it remains there forever.

Physical Interpretation An equilibrium represents a steady state of the system. Examples include

- A pendulum hanging vertically downward.
- A chemical reaction that has reached equilibrium.
- Constant population in an ecological model.
- Constant temperature in a thermal system.

The important question is not only whether an equilibrium exists, but also

Will the system remain close to the equilibrium if it is slightly disturbed?

This leads to the concept of **stability**.

Example: A simple pendulum

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0,$$

where θ is the angular displacement, g is the acceleration due to gravity, and L is the length of the pendulum. This can be expressed as

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -\frac{g}{L} \sin x_1, \end{aligned}$$

where $x_1 = \theta$ and $x_2 = \dot{\theta}$. The simple pendulum possesses two distinct equilibrium positions in the interval $0 \leq \theta < 2\pi$.

1. Pendulum hanging downward

$$\theta = 0,$$

which is a **stable equilibrium**. Small disturbances produce oscillations around this position.

2. Pendulum balanced vertically upward

$$\theta = \pi,$$

which is an **unstable equilibrium**. Even a tiny disturbance causes the pendulum to fall away from this position.

Hence, two equilibrium points may exist in the same system with completely different stability properties.

Example: Linear Harmonic Oscillator The system is

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -\omega_0^2 x. \end{aligned}$$

Setting

$$\dot{x} = 0, \quad \dot{y} = 0,$$

gives

$$\begin{aligned} y &= 0, \\ x &= 0. \end{aligned}$$

Hence

$$X^* = (0, 0).$$

Example The Lorenz equations are

$$\dot{x} = \sigma(y - x),$$

$$\dot{y} = rx - y - xz,$$

$$\dot{z} = xy - bz.$$

Setting all derivatives equal to zero,

$$\sigma(y - x) = 0,$$

$$rx - y - xz = 0,$$

$$xy - bz = 0.$$

The equilibrium points are

$$(0, 0, 0),$$

and

$$\begin{aligned} & \left(\sqrt{b(r-1)}, \sqrt{b(r-1)}, r-1 \right), \\ & \left(-\sqrt{b(r-1)}, -\sqrt{b(r-1)}, r-1 \right). \end{aligned}$$

Thus the Lorenz system possesses three equilibrium points whenever $r > 1$.

2.6 Phase Space, Phase Plane, and Phase Trajectories

The qualitative study of a dynamical system is greatly simplified by examining its motion in a geometric space called the **phase space**. Rather than studying the variation of each state variable separately with time, the complete state of the system is represented by a single point whose motion describes the evolution of the system. Phase space analysis provides valuable information about the long-term behavior of dynamical systems, including equilibrium points, periodic solutions, attractors, and stability.

Phase Space Consider an n -dimensional dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}),$$

where

$$\mathbf{X} = (x_1, x_2, \dots, x_n)^T$$

is the state vector. The n variables

$$(x_1, x_2, \dots, x_n)$$

serve as the coordinate axes of an n -dimensional **phase space**. Each point in this space uniquely represents the instantaneous state of the system. As time evolves, the point moves continuously through the phase space, tracing out a curve called the **phase trajectory**.

Example Consider the system

$$\begin{aligned}\dot{x} &= -x, \\ \dot{y} &= -2y, \\ \dot{z} &= -3z.\end{aligned}$$

The state vector is

$$X = (x, y, z)^T.$$

Therefore, the phase space is three-dimensional with coordinate axes x , y , and z . For the initial condition

$$X(0) = (1, 1, 1)^T,$$

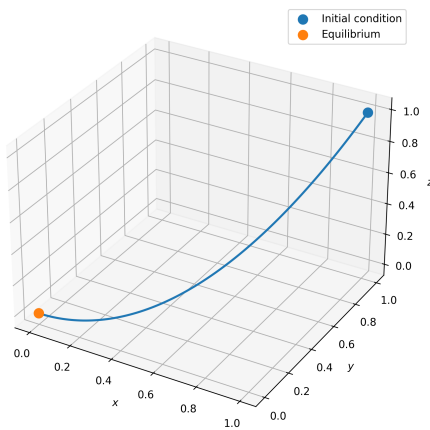
the solution is

$$x(t) = e^{-t}, \quad y(t) = e^{-2t}, \quad z(t) = e^{-3t}.$$

Hence, the phase point

$$X(t) = (e^{-t}, e^{-2t}, e^{-3t})^T$$

moves through the three-dimensional phase space and approaches the origin as $t \rightarrow \infty$.



Phase Plane If the system contains only two state variables,

$$\mathbf{X} = (x, y),$$

the phase space becomes two-dimensional and is called the **phase plane**. Typical examples include

$$\begin{aligned}\dot{x} &= f(x, y), \\ \dot{y} &= g(x, y).\end{aligned}$$

For second-order differential equations,

$$\ddot{x} = f(x, \dot{x}),$$

we usually define

$$y = \dot{x},$$

so that the phase plane consists of the variables

$$(x, y).$$

Each point on the phase plane represents the position and velocity of the system at a given instant.

Example: Simple Harmonic Oscillator Consider the second-order differential equation

$$\ddot{x} + x = 0.$$

Introducing the state variable

$$y = \dot{x},$$

the equation can be written as the first-order system

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -x.\end{aligned}$$

Thus, the phase plane is the (x, y) -plane, where x represents the position and y represents the velocity. For the initial condition

$$x(0) = 1, \quad y(0) = 0,$$

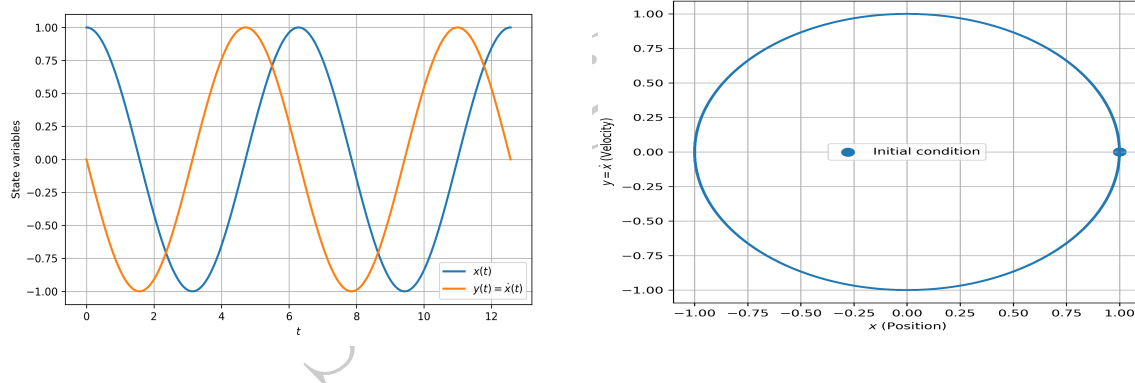
the solution is

$$x(t) = \cos t, \quad y(t) = -\sin t.$$

Eliminating t , we obtain

$$x^2 + y^2 = 1.$$

Therefore, the phase trajectory is a circle in the phase plane.



Phase Trajectories The curve traced by the moving phase point in the phase space is called the **phase trajectory** (or phase orbit). Mathematically,

$$\mathbf{X}(t) = (x_1(t), x_2(t), \dots, x_n(t))$$

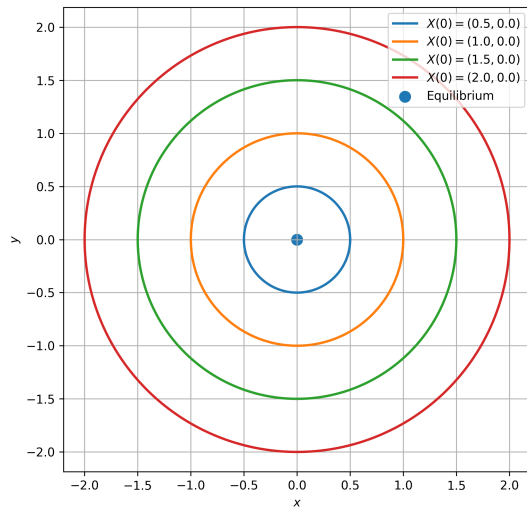
defines a trajectory in phase space. Different initial conditions generally generate different trajectories. Unlike ordinary graphs of $x(t)$ versus t , a phase trajectory directly illustrates the geometric behavior of the dynamical system. Now the quantity

$$x^2 + y^2$$

remains constant along every trajectory. Therefore,

$$x^2 + y^2 = C,$$

where the value of C depends on the initial condition. Hence, different initial conditions generate different circular trajectories in the phase plane. Each circle represents a different motion of the system.



2.7 Stability, Attractors and Repellers

Equilibrium Points in Phase Space Consider the dynamical system

$$\dot{X} = F(X).$$

An equilibrium point X^* satisfies

$$F(X^*) = 0.$$

In phase space, X^* appears as a fixed point because its coordinates do not change with time. If the system starts exactly at the equilibrium point,

$$X(0) = X^*,$$

then

$$X(t) = X^* \quad \forall t \geq 0.$$

Example Consider the system

$$\dot{X} = -X.$$

The equilibrium point is obtained from

$$-X^* = 0,$$

which gives

$$X^* = 0.$$

If

$$X(0) = 0,$$

then the solution is

$$X(t) = 0 \quad \forall t \geq 0.$$

Thus, the origin appears as a fixed point in phase space.

Behavior Near Equilibrium Points Knowing the location of an equilibrium point alone is not sufficient to understand the behavior of a dynamical system. We must also examine what happens to trajectories that start close to the equilibrium.

Three typical behaviors are illustrated below.

Case 1: Asymptotically Stable Equilibrium If trajectories starting sufficiently close to an equilibrium remain close and eventually approach it as $t \rightarrow \infty$, the equilibrium is called *asymptotically stable*.

Example Consider

$$\dot{X} = -X,$$

where $X = (x, y)$. The solution corresponding to the initial condition $X(0) = X_0$ is

$$X(t) = X_0 e^{-t}.$$

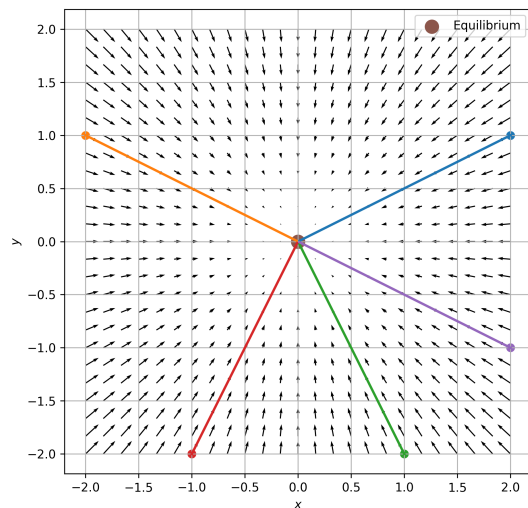
Therefore,

$$\lim_{t \rightarrow \infty} X(t) = 0.$$

Hence, the equilibrium

$$X^* = 0$$

is asymptotically stable. Nearby trajectories are attracted toward the origin.



Case 2: Unstable Equilibrium If trajectories starting arbitrarily close to an equilibrium move away from it, the equilibrium is called *unstable*.

Example Consider

$$\dot{X} = X,$$

where $X = (x, y)$. Its solution is

$$X(t) = X_0 e^t.$$

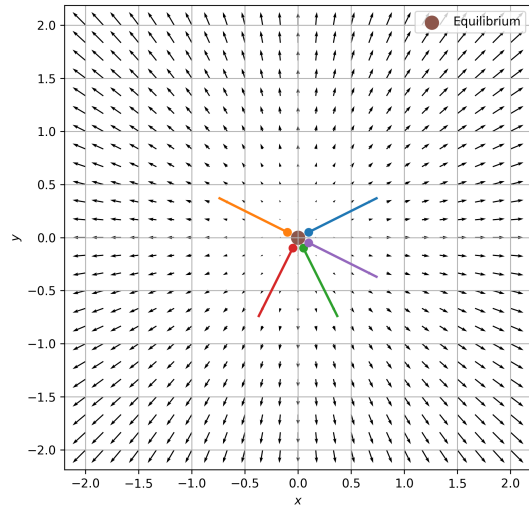
For every nonzero initial condition X_0 ,

$$|X(t)| \rightarrow \infty \quad \text{as } t \rightarrow \infty.$$

Thus, the equilibrium

$$X^* = 0$$

is unstable. Nearby trajectories are repelled from the origin.



Case 3: Neutrally Stable Equilibrium An equilibrium may be stable without attracting nearby trajectories. In this case, nearby trajectories remain close to the equilibrium but do not converge to it.

Example Consider the harmonic oscillator

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= -x.\end{aligned}$$

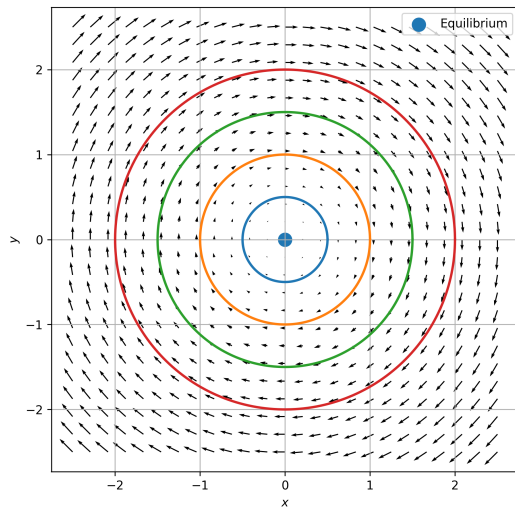
The quantity

$$x^2 + y^2$$

remains constant along every trajectory. Hence,

$$x^2 + y^2 = C.$$

The phase trajectories are circles centered at the origin. Trajectories neither approach nor move away from the equilibrium. Therefore, the origin is stable but not asymptotically stable. Such behavior is often referred to as neutral stability.



Definition 3. An equilibrium point X^* is called asymptotically stable if it is stable and trajectories starting sufficiently close to X^* satisfy

$$\lim_{t \rightarrow \infty} \|X(t) - X^*\| = 0.$$

Thus, the distance between the trajectory and the equilibrium tends to zero as time increases.

2.7.1 Attractors

An attractor is a set in phase space toward which nearby trajectories converge as time increases.

Definition 4. A set A in phase space is called an attractor if trajectories starting from a suitable neighborhood of A approach the set as $t \rightarrow \infty$. Symbolically,

$$\lim_{t \rightarrow \infty} \text{dist}(X(t), A) = 0.$$

Examples of attractors include asymptotically stable equilibrium points, and stable limit cycles.

Example Consider a damped simple pendulum governed by

$$\begin{aligned}\dot{\theta} &= \omega, \\ \dot{\omega} &= -c\omega - \frac{g}{L} \sin \theta.\end{aligned}$$

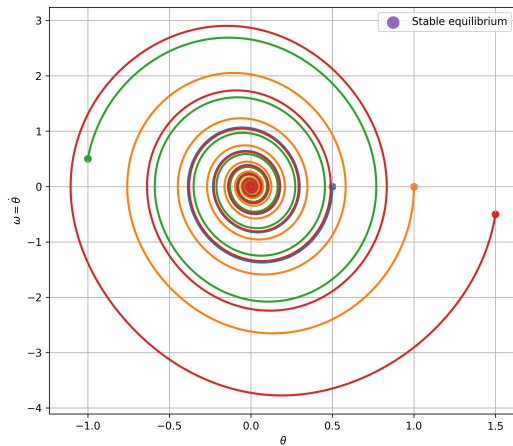
The downward vertical position

$$(\theta, \omega) = (0, 0)$$

is an equilibrium point. If the pendulum is slightly displaced, it oscillates about the downward position. Because of damping, energy is continuously dissipated and the amplitude of oscillation gradually decreases. In the phase plane, the trajectory spirals inward toward

$$(0, 0).$$

Hence, the downward equilibrium is an attractor.



2.7.2 Repellers

A repeller has behavior opposite to that of an attractor.

Definition 5. A repeller is a point or invariant set from which nearby trajectories move away as time increases.

Example Consider

$$\dot{X} = X.$$

The origin

$$X^* = 0$$

is an equilibrium point. However, for $X_0 \neq 0$,

$$X(t) = X_0 e^t.$$

Thus, even a small perturbation from the origin grows exponentially. The origin therefore acts as a repeller.

2.7.3 Limit Cycles as Attractors

An attractor need not be an equilibrium point. A closed periodic trajectory may also attract nearby trajectories. Such an isolated closed trajectory is called a *limit cycle*.

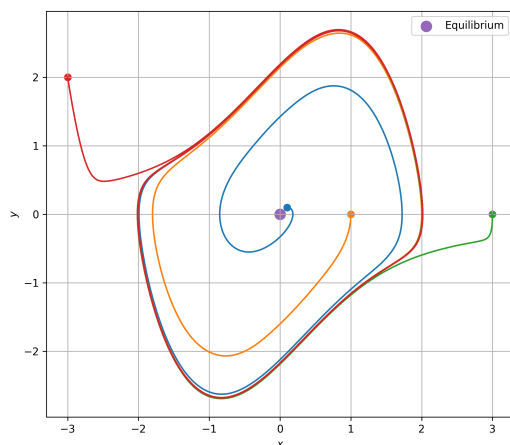
Example The Van der Pol oscillator

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \quad \mu > 0.$$

This can be rewritten as follows:

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= \mu(1 - x^2)y - x. \end{aligned}$$

For many initial conditions, the trajectories approach a closed periodic orbit. This closed orbit is a stable limit cycle and acts as an attractor.



How do trajectories approach a limit cycle? Consider two trajectories with different initial conditions: one starting inside the closed orbit and another starting outside the closed orbit. The trajectory starting inside the closed orbit gradually moves outward, whereas the trajectory starting outside the closed orbit gradually moves inward. Despite having different initial conditions, both trajectories eventually approach the same closed periodic orbit. Thus, the long-term behavior of the system becomes independent of these initial conditions. The attracting set is not an equilibrium point but a closed periodic trajectory called a *stable limit cycle*. This illustrates an important principle: An attractor need not be a point. An attractor may be an equilibrium point, a periodic orbit such as a stable limit cycle, or a more complicated invariant set.

2.8 Classification of Equilibrium Points: Two-Dimensional Systems

In the previous section, we introduced the concept of equilibrium (fixed) points and discussed their qualitative behavior. In this section, we develop a mathematical criterion to determine whether an equilibrium point is **stable**, **unstable**, or **neutrally stable**. The method is known as **linear stability analysis**. The basic idea is to introduce a very small perturbation around an equilibrium point and study the evolution of the perturbation with time. If the perturbation decays, the equilibrium is stable; if it grows, the equilibrium is unstable.

General Two-Dimensional Dynamical System Consider the autonomous system

$$\begin{aligned}\dot{x} &= P(x, y), \\ \dot{y} &= Q(x, y),\end{aligned}$$

where $P(x, y)$ and $Q(x, y)$ are continuously differentiable functions. Let

$$X^* = (x_0, y_0)$$

be an equilibrium point. Then

$$P(x_0, y_0) = 0, \quad Q(x_0, y_0) = 0.$$

Since both derivatives vanish at this point, the state of the system does not change with time. To study stability, we slightly disturb the equilibrium point by introducing small perturbations

$$x = x_0 + \xi(t), \quad y = y_0 + \eta(t),$$

where

$$|\xi|, |\eta| \ll 1.$$

The functions $\xi(t)$ and $\eta(t)$ represent small deviations from the equilibrium. Our objective is to determine whether these perturbations decrease or increase with time. Expand the functions $P(x, y)$ and $Q(x, y)$ about the equilibrium point. Using the first-order Taylor expansion,

$$P(x, y) \approx P(x_0, y_0) + \left. \frac{\partial P}{\partial x} \right|_{X^*} \xi + \left. \frac{\partial P}{\partial y} \right|_{X^*} \eta,$$

$$Q(x, y) \approx Q(x_0, y_0) + \left. \frac{\partial Q}{\partial x} \right|_{X^*} \xi + \left. \frac{\partial Q}{\partial y} \right|_{X^*} \eta.$$

Since

$$P(x_0, y_0) = Q(x_0, y_0) = 0,$$

the constant terms vanish. Ignoring quadratic and higher-order terms gives the linear approximation

$$\dot{\xi} = a\xi + b\eta,$$

$$\dot{\eta} = c\xi + d\eta,$$

where

$$a = \left. \frac{\partial P}{\partial x} \right|_{X^*}, \quad b = \left. \frac{\partial P}{\partial y} \right|_{X^*}, \quad c = \left. \frac{\partial Q}{\partial x} \right|_{X^*}, \quad d = \left. \frac{\partial Q}{\partial y} \right|_{X^*}.$$

Thus, the nonlinear system is approximated by a linear system in a small neighborhood of the equilibrium. The coefficients of the linearized system form the **Jacobian matrix**

$$J(X^*) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The Jacobian contains all first-order partial derivatives evaluated at the equilibrium point. The local behavior of the nonlinear system depends entirely on this matrix. Assume a solution of the form

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \mathbf{v}e^{\lambda t},$$

where $\mathbf{v}$ is a constant vector. Substituting into the linearized equations gives

$$(J - \lambda I)\mathbf{v} = 0.$$

For a nontrivial solution,

$$\det(J - \lambda I) = 0.$$

Expanding,

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0.$$

Hence the eigenvalues are

$$\lambda_{1,2} = \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - bc)}}{2}.$$

Define

$$T = a + d,$$

called the **trace** of the Jacobian, and

$$D = ad - bc,$$

called the **determinant** of the Jacobian. The characteristic equation becomes

$$\lambda^2 - T\lambda + D = 0.$$

The eigenvalues are therefore

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4D}}{2}.$$

Role of the Trace, Determinant, and Discriminant The classification of an equilibrium point can be understood from three quantities:

$$T = \text{tr}(J), \quad D = \det(J), \quad \Delta = T^2 - 4D.$$

The trace determines the sum of the eigenvalues, $\lambda_1 + \lambda_2 = T$, while the determinant determines their product, $\lambda_1 \lambda_2 = D$. The discriminant $\Delta = T^2 - 4D$ determines whether the eigenvalues are real or complex.

- If $D < 0$, the eigenvalues have opposite signs. Therefore, the equilibrium is a **saddle point** and is always unstable.
- If $D > 0$, $\Delta > 0$, the eigenvalues are real and have the same sign. The equilibrium is a **node**. The sign of T determines whether the node is stable or unstable.
- If $D > 0$, $\Delta < 0$, the eigenvalues are complex conjugates. The equilibrium is a **focus**. The sign of T determines whether trajectories spiral inward or outward.
- If $D > 0$, $T = 0$, the eigenvalues are purely imaginary. For a linear system, the equilibrium is a **center**.
- If $D = 0$, at least one eigenvalue is zero. The equilibrium is non-hyperbolic, and linearization may be inconclusive.

Thus, the quantities T , D , and Δ provide a convenient method for classifying equilibrium points of two-dimensional systems.

Condition on Eigenvalues	Local Behavior
$\Re(\lambda_1) < 0, \Re(\lambda_2) < 0$	Asymptotically stable
$\Re(\lambda_1) > 0$ or $\Re(\lambda_2) > 0$ or $\Re(\lambda_1), \Re(\lambda_2) > 0$	Unstable
Eigenvalues have opposite signs	Saddle point; unstable
At least one eigenvalue has zero real part	Linearization may be inconclusive

Interpretation If $\Re(\lambda_1) < 0$, $\Re(\lambda_2) < 0$, all sufficiently small perturbations decay exponentially. Hence, nearby trajectories approach the equilibrium point. If at least one eigenvalue satisfies $\Re(\lambda) > 0$, there exists a perturbation that grows exponentially. Therefore, the equilibrium is unstable. If one or more eigenvalues have zero real part, the equilibrium is *non-hyperbolic*. In this case, linearization may not determine the stability of the original nonlinear system, and higher-order nonlinear terms must be examined. For a purely linear system, purely imaginary eigenvalues produce a center and closed trajectories. For a nonlinear system, however, purely imaginary eigenvalues alone do not guarantee neutral stability.

Linearization Linearization provides an accurate approximation of the nonlinear system only in a sufficiently small neighborhood of the equilibrium point. Far from the equilibrium, neglected nonlinear terms become significant and may alter the behavior of the system. Therefore, linear stability analysis is a **local** stability analysis.

Classification of Equilibrium Points The eigenvalues of the Jacobian determine the geometry of nearby phase trajectories. The principal types of equilibrium points are described below.

1. Stable Node (Sink) Suppose

$$\lambda_1 < 0, \quad \lambda_2 < 0, \quad \lambda_1 \neq \lambda_2.$$

Both eigenvalues are real and negative. Therefore, all nearby trajectories approach the equilibrium point as $t \rightarrow \infty$. The two eigenvalues generally correspond to different rates of exponential decay. Consequently, trajectories approach the equilibrium preferentially along the direction associated with the eigenvalue having the smaller magnitude. The equilibrium is asymptotically stable and acts as a sink.

2. Unstable Node (Source) Suppose

$$\lambda_1 > 0, \quad \lambda_2 > 0, \quad \lambda_1 \neq \lambda_2.$$

Both eigenvalues are real and positive. Perturbations grow exponentially and nearby trajectories move away from the equilibrium. The equilibrium is unstable and acts as a source.

3. Stable Focus (Spiral Sink) Suppose

$$\lambda_{1,2} = \alpha \pm i\beta, \quad \alpha < 0, \quad \beta \neq 0.$$

The imaginary part produces oscillatory motion, while the negative real part causes the amplitude of oscillation to decay exponentially. Hence, trajectories spiral inward toward the equilibrium. The equilibrium is asymptotically stable and is called a spiral sink or stable focus.

4. Unstable Focus (Spiral Source) Suppose

$$\lambda_{1,2} = \alpha \pm i\beta, \quad \alpha > 0, \quad \beta \neq 0.$$

The imaginary part produces oscillations, while the positive real part causes their amplitude to increase. Thus, trajectories spiral outward from the equilibrium. The equilibrium is unstable and is called a spiral source or unstable focus.

5. Center For a linear system, suppose

$$\lambda_{1,2} = \pm i\beta, \quad \beta \neq 0.$$

There is no exponential growth or decay. The phase trajectories are closed curves surrounding the equilibrium point. The equilibrium is stable but not asymptotically stable. For nonlinear systems, purely imaginary eigenvalues make the equilibrium non-hyperbolic. Linearization alone may not determine whether the equilibrium is a center, stable, or unstable.

6. Saddle Point Suppose

$$\lambda_1 < 0, \quad \lambda_2 > 0.$$

The eigenvalues are real and have opposite signs. Perturbations decay along the eigendirection corresponding to the negative eigenvalue and grow along the eigendirection corresponding to the positive eigenvalue. Thus, trajectories approach the equilibrium along one direction and move away along another direction. A saddle point is always unstable.

7. Repeated Eigenvalues Suppose

$$\lambda_1 = \lambda_2 = \lambda.$$

The nature of the equilibrium depends not only on the repeated eigenvalue but also on the number of linearly independent eigenvectors. If the Jacobian has two linearly independent eigenvectors, the equilibrium is called a *star node*. Trajectories move along radial directions. If the Jacobian has only one linearly independent eigenvector, the equilibrium is called an *improper node*. Trajectories become tangent to the single eigendirection near the equilibrium. For

$$\lambda < 0,$$

the equilibrium is asymptotically stable, whereas for

$$\lambda > 0,$$

the equilibrium is unstable.

8. Degenerate or Non-Hyperbolic Equilibrium An equilibrium is called non-hyperbolic if at least one eigenvalue has zero real part. An important case occurs when

$$\det(J) = 0,$$

so that at least one eigenvalue is zero. In such cases, linearization may be insufficient to determine the local behavior. Higher-order nonlinear terms may significantly influence the dynamics. Techniques such as nonlinear stability analysis, Lyapunov methods, or center manifold theory may be required.

Example Consider the linear system

$$\dot{X} = AX.$$

The following examples illustrate the principal types of equilibrium points.

1. Stable Node

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = -1, \quad \lambda_2 = -2.$$

Hence, the origin is a stable node.

2. Unstable Node

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2.$$

Hence, the origin is an unstable node.

3. Stable Focus

$$A = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_{1,2} = -1 \pm i.$$

Hence, the origin is a stable focus.

4. Unstable Focus

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_{1,2} = 1 \pm i.$$

Hence, the origin is an unstable focus.

5. Center

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_{1,2} = \pm i.$$

Hence, the origin is a center.

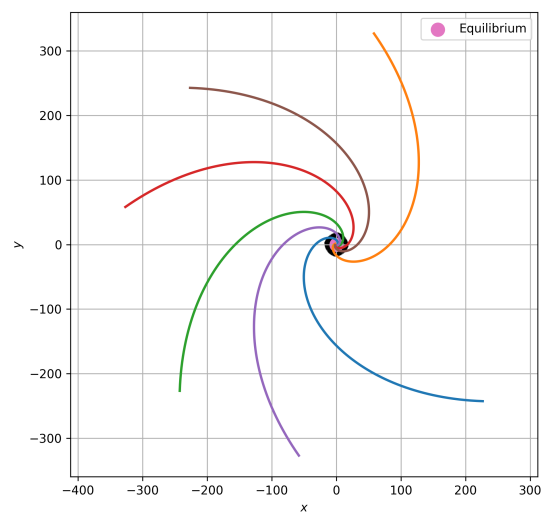
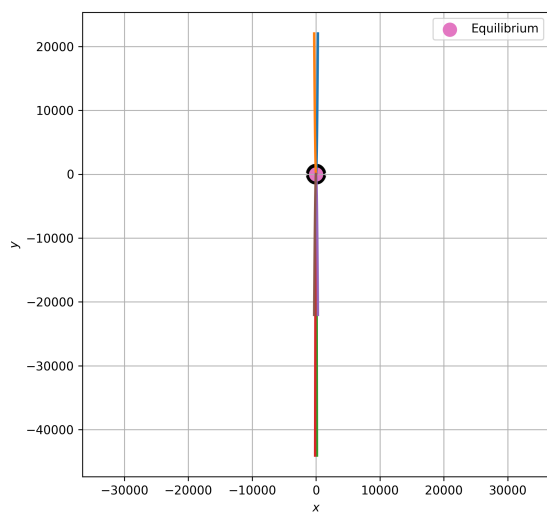
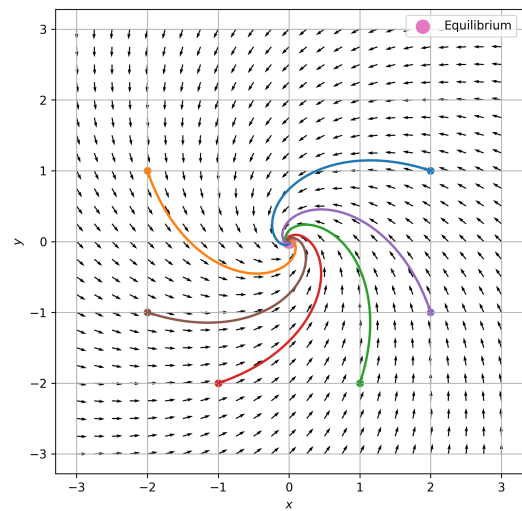
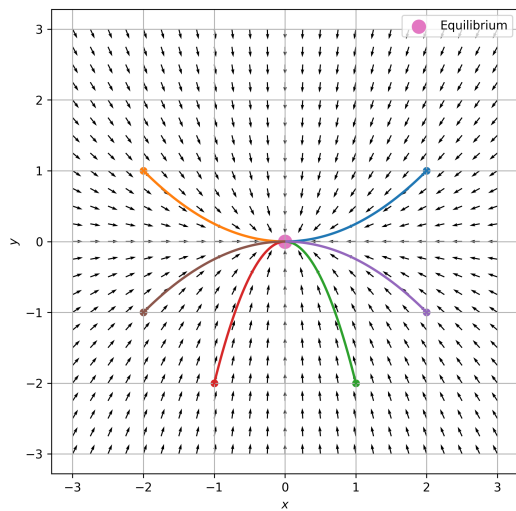
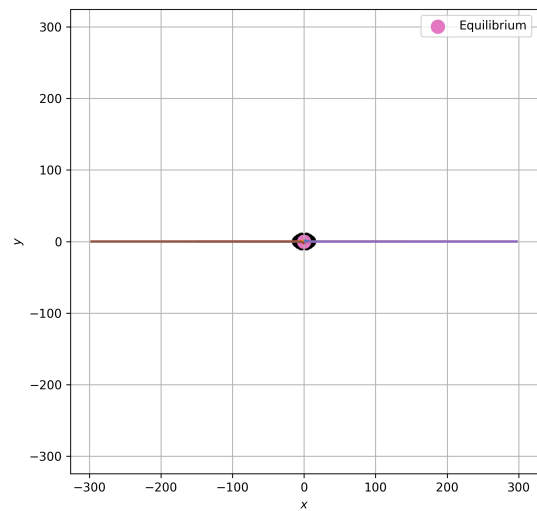
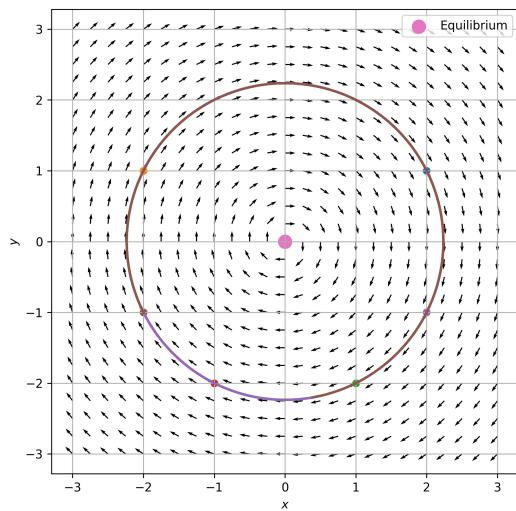
6. Saddle Point

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = -1.$$

Hence, the origin is a saddle point.



Geometric Interpretation of Eigenvalues The eigenvalues determine whether perturbations grow or decay, while the eigenvectors determine the preferred directions of motion in the phase plane. For real eigenvalues, trajectories tend to align with the eigendirections. A negative eigenvalue produces decay along its eigendirection, whereas a positive eigenvalue produces growth. Complex eigenvalues produce rotational behavior. If

$$\lambda = \alpha \pm i\beta,$$

the real part α determines the change in amplitude, while the imaginary part β produces rotation or oscillation. Thus,

$$\begin{aligned}\Re(\lambda) & \text{ determines growth or decay,} \\ \Im(\lambda) & \text{ determines oscillation or rotation.}\end{aligned}$$

This provides a direct connection between the algebraic properties of the Jacobian matrix and the geometric structure of trajectories in the phase plane.

Trace-Determinant Plane The classification of a two-dimensional equilibrium point can be represented geometrically in the trace-determinant plane. The curve

$$D = \frac{T^2}{4}$$

corresponds to

$$\Delta = T^2 - 4D = 0$$

and separates regions with real eigenvalues from regions with complex eigenvalues. Below this curve, when

$$D > 0,$$

the eigenvalues are real, and the equilibrium is a node. Above the curve, the eigenvalues are complex conjugates and the equilibrium is a focus. The line

$$D = 0$$

separates saddle points from nodes and foci. If

$$D < 0,$$

the eigenvalues have opposite signs, and the equilibrium is always a saddle point. Finally, the sign of the trace determines attraction or repulsion:

$$T < 0 \implies \text{attracting behavior,}$$

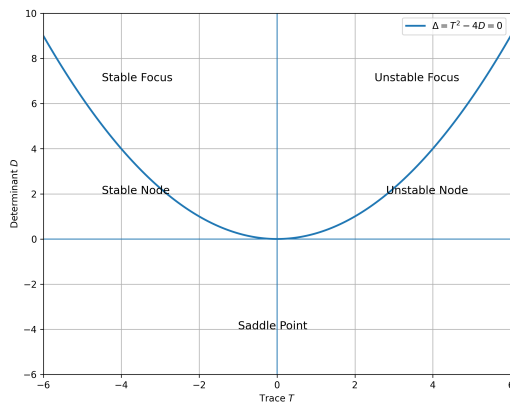
whereas

$$T > 0 \implies \text{repelling behavior.}$$

Thus, the location of the point

$$(T, D)$$

in the trace-determinant plane provides a rapid geometric classification of the equilibrium point.



Example: Damped Cubic Oscillator Consider

$$\ddot{x} + d\dot{x} - ax + bx^3 = 0,$$

where

$$a > 0, \quad b > 0, \quad d > 0.$$

Thus

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -dy + ax - bx^3. \end{aligned}$$

The equilibrium points are

$$(0, 0), \quad \left(\sqrt{\frac{a}{b}}, 0\right), \quad \left(-\sqrt{\frac{a}{b}}, 0\right).$$

The Jacobian matrix is

$$J = \begin{bmatrix} 0 & 1 \\ a - 3bx^2 & -d \end{bmatrix}.$$

From the eigenvalues we have

At the origin

$$J(0, 0) = \begin{bmatrix} 0 & 1 \\ a & -d \end{bmatrix}.$$

The determinant is

$$D = -a < 0.$$

Therefore, the eigenvalues have opposite signs, and the origin is a **saddle point**. Hence, the equilibrium

$$(0, 0)$$

is unstable.

At the equilibria

$$\left(\pm\sqrt{\frac{a}{b}}, 0\right),$$

we have

$$a - 3bx^2 = a - 3a = -2a.$$

Thus, the Jacobian becomes

$$J = \begin{bmatrix} 0 & 1 \\ -2a & -d \end{bmatrix}.$$

The trace and determinant are

$$T = -d < 0, \quad D = 2a > 0.$$

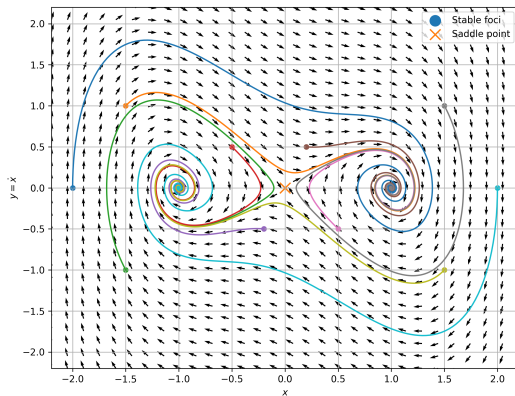
The discriminant is

$$\Delta = T^2 - 4D = d^2 - 8a.$$

Therefore, three cases arise:

$$\begin{aligned} d^2 > 8a &\implies \text{stable node,} \\ d^2 = 8a &\implies \text{repeated negative eigenvalue,} \\ d^2 < 8a &\implies \text{stable focus.} \end{aligned}$$

Thus, the same nonlinear dynamical system may contain equilibrium points of different types. The origin is always a saddle point, whereas the two outer equilibria are asymptotically stable. Their precise geometric classification depends on the damping parameter d .



Interpretation of the Phase Portrait For

$$a = 1, \quad b = 1, \quad d = 0.5,$$

the equilibrium points are

$$(-1, 0), \quad (0, 0), \quad (1, 0).$$

At the origin,

$$D = -a = -1 < 0.$$

Hence, the origin is a saddle point. Trajectories approach the origin along one direction and move away along another direction. At the two outer equilibria,

$$(-1, 0) \quad \text{and} \quad (1, 0),$$

the discriminant is

$$\Delta = d^2 - 8a = (0.5)^2 - 8 = -7.75 < 0.$$

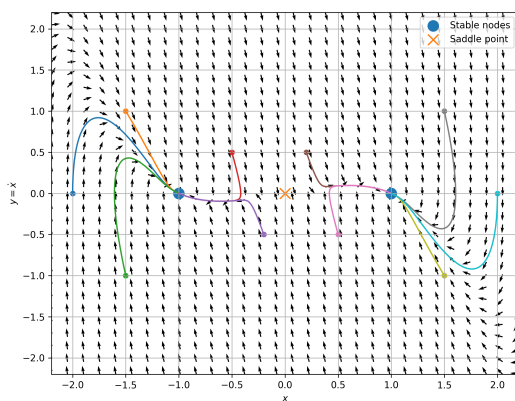
Therefore, the eigenvalues are complex conjugates with negative real parts. Consequently, the two outer equilibria are stable foci. The phase portrait shows trajectories spiraling toward one of the two stable equilibria. The long-term state of the system depends on the initial condition. Some trajectories approach

$$(-1, 0),$$

whereas others approach

$$(1, 0).$$

Thus, the system possesses two attracting equilibrium points separated by the saddle structure associated with the origin.



2.9 Limit Cycle Motion: Periodic Attractors

In the previous sections, we studied equilibrium points as possible long-term states of dynamical systems. For an asymptotically stable equilibrium, nearby trajectories eventually approach a fixed point in phase space. However, many dynamical systems do not settle to a state of rest. Instead, they approach a sustained periodic motion. In the phase plane, such periodic motion may appear as an isolated closed trajectory called a **limit cycle**. A stable limit cycle represents a **periodic attractor**.

Closed Orbits and Limit Cycles Consider a two-dimensional autonomous system

$$\begin{aligned}\dot{x} &= P(x, y), \\ \dot{y} &= Q(x, y).\end{aligned}$$

Suppose a solution satisfies

$$X(t + T) = X(t)$$

for some

$$T > 0.$$

The motion is periodic with period T . In the phase plane, the corresponding trajectory forms a closed curve. However, not every closed trajectory is a limit cycle.

Definition 6. A **limit cycle** is an isolated closed trajectory in the phase plane.

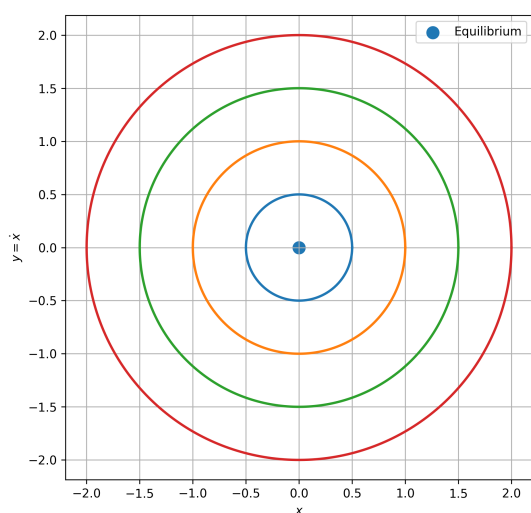
The word *isolated* means that sufficiently close to the closed trajectory, there are no other closed trajectories arbitrarily near it. For example, the ideal harmonic oscillator

$$\ddot{x} + x = 0$$

has the phase trajectories

$$x^2 + y^2 = C.$$

These trajectories form a continuous family of closed curves. Therefore, an individual circular trajectory is **not** a limit cycle. A limit cycle, in contrast, is an isolated periodic orbit.



The phase portrait of the harmonic oscillator contains a continuous family of closed trajectories. For every closed orbit, another closed orbit can be found arbitrarily close to it. Therefore, no individual periodic orbit is isolated. Hence, Closed orbit $\nRightarrow$ Limit cycle. A limit cycle must be an isolated closed trajectory.

Stable and Unstable Limit Cycles The behavior of nearby trajectories determines the stability of a limit cycle.

Stable Limit Cycle If trajectories starting sufficiently close to a closed orbit approach it as $t \rightarrow \infty$, the closed orbit is called a **stable limit cycle**. Thus, Stable Limit Cycle $\Rightarrow$ Periodic Attractor.

Unstable Limit Cycle If nearby trajectories move away from the closed orbit, the orbit is called an **unstable limit cycle**. An unstable limit cycle acts as a periodic repeller.

Example: Stable Limit Cycle Consider the system in polar coordinates

$$\begin{aligned}\dot{r} &= r(1 - r^2), \\ \dot{\theta} &= 1.\end{aligned}$$

The equilibrium points are $r = 0$ and $r = 1$. The angular equation

$$\dot{\theta} = 1$$

shows that every trajectory rotates around the origin with constant angular velocity. To determine the nature of the circle $r = 1$, consider three cases.

- If

$$0 < r < 1,$$

then

$$\dot{r} = r(1 - r^2) > 0,$$

so the radius increases with time. The trajectory spirals outward toward the unit circle.

- If

$$r > 1,$$

then

$$\dot{r} = r(1 - r^2) < 0,$$

so the radius decreases with time. The trajectory spirals inward toward the unit circle.

- If

$$r = 1,$$

then

$$\dot{r} = 0,$$

and the trajectory remains on the circle of radius one.

Hence, trajectories starting both inside and outside the unit circle approach the same closed orbit. Therefore,

$$r = 1$$

is a **stable limit cycle**. The unit circle acts as a **periodic attractor**. The origin,

$$r = 0,$$

is an unstable equilibrium point, since a small perturbation causes the trajectory to move away from the origin and eventually approach the stable limit cycle.

Example: Unstable Limit Cycle Consider the system in polar coordinates

$$\begin{aligned}\dot{r} &= r(r-1), \\ \dot{\theta} &= 1.\end{aligned}$$

The equilibrium solutions of the radial equation satisfy $r(r-1) = 0$, giving $r = 0$ and $r = 1$. The angular equation

$$\dot{\theta} = 1$$

shows that every trajectory rotates around the origin with constant angular velocity. To determine the nature of the circle $r = 1$, consider three cases.

- If $0 < r < 1$,

$$\dot{r} = r(r-1) < 0,$$

so the radius decreases, and trajectories spiral inward toward the origin.

- If $r > 1$,

$$\dot{r} = r(r-1) > 0,$$

so the radius increases, and trajectories spiral outward away from the unit circle.

- If $r = 1$,

$$\dot{r} = 0,$$

and the trajectory remains on the circle of radius one.

Hence, trajectories starting on either side of the circle move away from it. Therefore,

$$r = 1$$

is an **unstable limit cycle**. The origin

$$r = 0$$

is a stable equilibrium point.

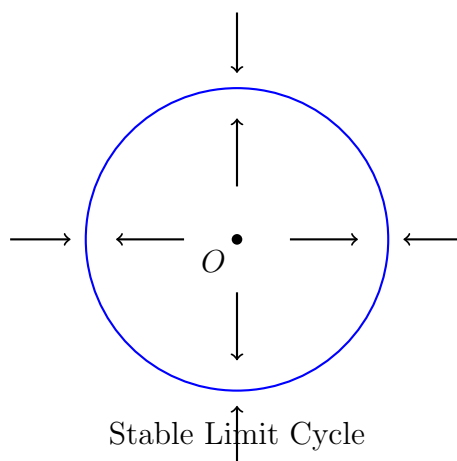


Figure 3: Trajectories starting inside the unit circle spiral outward, whereas those starting outside spiral inward. Both approach the same periodic orbit, making the unit circle a stable limit cycle (periodic attractor).

Physical Example: Wall-Clock Pendulum In a wall-clock pendulum. Two competing effects occur:

1. air resistance and friction continuously remove energy from the pendulum
2. the escapement mechanism periodically supplies energy to the system.

If the energy supplied by the escapement balances the energy dissipated by friction, the pendulum eventually exhibits oscillations of approximately constant amplitude. The long-term motion is periodic and is qualitatively similar to an attracting limit cycle. This suggests an important mechanism for sustained oscillations and can produce a stable periodic motion.

The Van der Pol Oscillator One of the classical mathematical models exhibiting stable limit-cycle behavior is the Van der Pol oscillator. Its equation can be written as the first-order system

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= b(1 - x^2)y - x.\end{aligned}$$

The only equilibrium point is $(0, 0)$. The Jacobian matrix is

$$J(x, y) = \begin{bmatrix} 0 & 1 \\ -2bxy - 1 & b(1 - x^2) \end{bmatrix}.$$

At the origin,

$$J(0, 0) = \begin{bmatrix} 0 & 1 \\ -1 & b \end{bmatrix}.$$

The characteristic equation is

$$\det(J - \lambda I) = 0,$$

which gives

$$\lambda^2 - b\lambda + 1 = 0.$$

Hence,

$$\lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4}}{2}.$$

For $b > 0$, the trace is $T = b > 0$, and the determinant is $D = 1 > 0$. Therefore, the equilibrium at the origin is unstable. For

$$0 < b$$

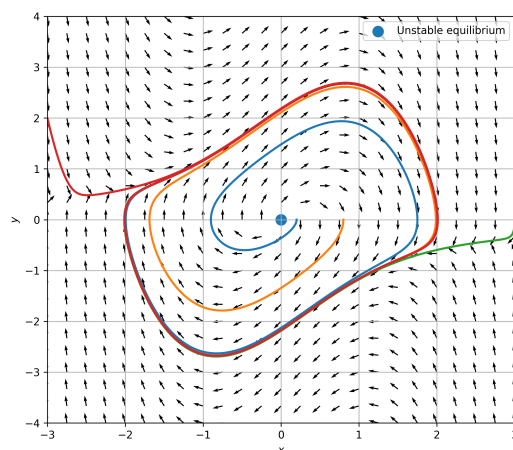
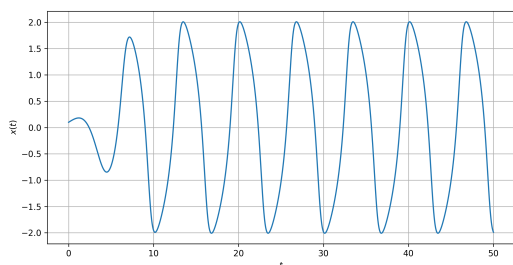
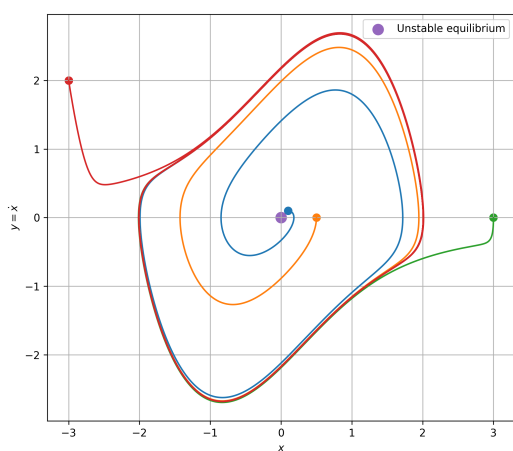
the eigenvalues are complex conjugates with positive real parts, and the origin is an unstable focus. Thus, trajectories starting close to the origin initially move away from the equilibrium.

Phase-Plane Interpretation The phase portrait of the Van der Pol oscillator reveals the mechanism of the periodic attractor. Consider two trajectories:

1. one starting inside the stable limit cycle
2. another starting outside the stable limit cycle.

The trajectory starting inside the closed orbit moves outward because small oscillations gain energy. The trajectory starting outside the closed orbit moves inward because large oscillations lose energy. Eventually, both trajectories approach the same closed periodic orbit. Thus, an attractor need not be a point. In this example, the attracting set is a closed periodic trajectory rather than an equilibrium point.

Transient and Steady Periodic Motion Before reaching the limit cycle, the trajectory undergoes transient motion. During this stage, the oscillation amplitude changes with time. As $t \rightarrow \infty$, the transient behavior disappears, and the trajectory approaches the stable limit cycle. The resulting motion is periodic $X(t + T) = X(t)$. Thus, the long-term behavior of the system is a sustained oscillation with a well-defined period and amplitude. The stable limit cycle therefore acts as a Periodic Attractor.



2.10 Poincaré-Bendixson Theorem

Theorem 1 (Bendixson's Criterion). *Consider*

$$\dot{x} = P(x, y), \quad \dot{y} = Q(x, y),$$

where

$$P, Q \in C^1(D).$$

Define the divergence

$$s(x, y) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}.$$

If $s(x, y)$ does not change sign and is not identically zero throughout a simply connected region D , then no closed periodic orbit lies completely inside D .

(The quantity

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$$

measures the rate of expansion or contraction of phase-space area.

- $\nabla \cdot F < 0$ indicates area contraction.
- $\nabla \cdot F > 0$ indicates area expansion.
- If the divergence keeps the same sign throughout a region, closed periodic motion cannot persist inside that region.)

Proof. Assume, contrary to the statement, that a periodic orbit C exists inside the region D . Let R denote the region enclosed by C . By Green's theorem,

$$\iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA = \oint_C (P dy - Q dx).$$

Since C is a trajectory of the differential equation,

$$dx = P dt, \quad dy = Q dt.$$

Therefore,

$$P dy - Q dx = PQ dt - QP dt = 0.$$

Hence

$$\iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA = 0.$$

However, if

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$$

has the same nonzero sign throughout R , its integral cannot be zero. This contradiction proves that no periodic orbit can exist. $\square$

A System with a Stable Limit Cycle Consider the nonlinear system in polar coordinates

$$\begin{aligned}\dot{r} &= r(1 - r^2), \\ \dot{\theta} &= 1.\end{aligned}$$

This system possesses a stable limit cycle at

$$r = 1.$$

To examine Bendixson's Criterion, we first write the system in Cartesian coordinates:

$$\begin{aligned}\dot{x} &= x(1 - r^2) - y, \\ \dot{y} &= y(1 - r^2) + x,\end{aligned}$$

where

$$r^2 = x^2 + y^2.$$

Thus,

$$P(x, y) = x(1 - r^2) - y,$$

and

$$Q(x, y) = y(1 - r^2) + x.$$

The divergence is

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2 - 4(x^2 + y^2).$$

Equivalently,

$$\nabla \cdot F = 2 - 4r^2.$$

Observe that

$$\nabla \cdot F > 0 \quad \text{when} \quad r < \frac{1}{\sqrt{2}},$$

whereas

$$\nabla \cdot F < 0 \quad \text{when} \quad r > \frac{1}{\sqrt{2}}.$$

Hence, the divergence changes sign within the plane. Since the hypothesis of Bendixson's Criterion is not satisfied, the theorem cannot be applied. Consequently, the existence of the stable limit cycle does not contradict Bendixson's Criterion. This example illustrates that Bendixson's Criterion provides only a sufficient condition for the nonexistence of periodic orbits. If the divergence changes sign, the criterion gives no conclusion regarding the existence or nonexistence of periodic solutions.

Application of Bendixson's Criterion Consider the linear system

$$\begin{aligned}\dot{x} &= -x, \\ \dot{y} &= -y.\end{aligned}$$

Here,

$$P(x, y) = -x, \quad Q(x, y) = -y.$$

Therefore,

$$\frac{\partial P}{\partial x} = -1, \quad \frac{\partial Q}{\partial y} = -1.$$

Hence, the divergence is

$$\nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -2.$$

Since

$$\nabla \cdot F = -2 < 0$$

throughout the entire plane, the divergence has a constant negative sign. Therefore, by Bendixson's Criterion, the system possesses no closed periodic orbit. Indeed, every trajectory satisfies

$$x(t) = x_0 e^{-t}, \quad y(t) = y_0 e^{-t},$$

which approaches the origin as $t \rightarrow \infty$. Thus, all trajectories converge to the stable equilibrium point at the origin, and no periodic motion is possible.

Damped Cubic Oscillator Consider

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -dy - x + x^3. \end{aligned}$$

Here, $P(x, y) = y$, and $Q(x, y) = -dy - x + x^3$. Therefore,

$$\frac{\partial P}{\partial x} = 0, \text{ and } \frac{\partial Q}{\partial y} = -d.$$

Hence,

$$\nabla \cdot F = -d.$$

Since $d > 0$, for the system, the divergence is always negative. It never changes sign. Therefore, by Bendixson's Criterion, No limit cycle exists.

Van der Pol Oscillator The state equations are

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= b(1 - x^2)y - x, \quad b > 0. \end{aligned}$$

The divergence is

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0 + b(1 - x^2).$$

Thus,

$$\nabla \cdot F = b(1 - x^2).$$

Since

$$\nabla \cdot F > 0, \quad |x| < 1,$$

and

$$\nabla \cdot F < 0, \quad |x| > 1,$$

the divergence changes sign. Hence Bendixson's Criterion cannot rule out periodic motion. Indeed, the Van der Pol oscillator possesses a stable limit cycle.

Bonhoeffer-Van der Pol Oscillator

$$\begin{aligned}\dot{x} &= x - \frac{x^3}{3} - y + A_0, \\ \dot{y} &= c(x + a - by).\end{aligned}$$

The divergence is

$$\nabla \cdot F = 1 - x^2 - bc.$$

Since this expression depends on x , it generally changes sign. Therefore, Bendixson's criterion is inconclusive. This system is well known to possess limit cycles for suitable parameter values.

Brusselator

$$\begin{aligned}\dot{x} &= a - (b+1)x + x^2y, \\ \dot{y} &= bx - x^2y, \quad a, b > 0\end{aligned}$$

The divergence is

$$\nabla \cdot F = (b-1) - 2xy.$$

Since the sign depends on (x, y) , Bendixson's criterion is inconclusive. The Brusselator is known to exhibit stable limit cycles for appropriate parameters.

Duffing-Van der Pol Oscillator

$$\ddot{x} - b(1 - x^2)\dot{x} + ax + \beta x^3 = 0.$$

State equations are

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= b(1 - x^2)y - ax - \beta x^3.\end{aligned}$$

Hence,

$$\nabla \cdot F = b(1 - x^2).$$

The divergence changes sign. Therefore, Bendixson's criterion cannot exclude limit cycles. Stable periodic oscillations occur for suitable parameters.

Example

$$\begin{aligned}\dot{x} &= y - x(x^2 + y^2 - Q), \\ \dot{y} &= -x - y(x^2 + y^2 - Q).\end{aligned}$$

The divergence is

$$\begin{aligned}\nabla \cdot F &= (-3x^2 - y^2 + Q) + (-x^2 - 3y^2 + Q) \\ &= 2Q - 4(x^2 + y^2).\end{aligned}$$

Since this changes sign depending on the distance from the origin, Bendixson's criterion is inconclusive. In fact, this system possesses $x^2 + y^2 = Q$ as a stable limit cycle whenever $Q > 0$.

2.11 Higher-Dimensional Dynamical Systems

The linear stability analysis developed for two-dimensional dynamical systems can be extended naturally to systems with n state variables. The procedure remains the same: locate the equilibrium points, linearize the nonlinear system about an equilibrium, and analyze the eigenvalues of the Jacobian matrix.

General n -Dimensional System Consider the autonomous dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}),$$

where

$$\mathbf{X} = (x_1, x_2, \dots, x_n)^T$$

is the state vector and

$$\mathbf{F} = (F_1, F_2, \dots, F_n)^T.$$

An equilibrium point satisfies

$$\mathbf{F}(\mathbf{X}^*) = 0.$$

To study the local behavior, introduce a small perturbation

$$\mathbf{X} = \mathbf{X}^* + \delta\mathbf{X}, \quad \|\delta\mathbf{X}\| \ll 1.$$

where

$$\delta\mathbf{X} = (\delta x_1, \delta x_2, \dots, \delta x_n)^T.$$

Expanding the vector field about the equilibrium point using the first-order Taylor series and neglecting higher-order terms gives

$$\delta\dot{\mathbf{X}} = \mathbf{A}\delta\mathbf{X},$$

where

$$\mathbf{A} = \left[\frac{\partial F_i}{\partial x_j} \right]_{\mathbf{X}=\mathbf{X}^*}$$

is called the Jacobian (stability) matrix evaluated at the equilibrium point. The solution of the linearized system is

$$\delta\mathbf{X}(t) = e^{\mathbf{A}t}\delta\mathbf{X}(0),$$

where

$$e^{\mathbf{A}t} = \mathbf{I} + t\mathbf{A} + \frac{t^2\mathbf{A}^2}{2!} + \frac{t^3\mathbf{A}^3}{3!} + \dots$$

is the matrix exponential. Suppose the Jacobian matrix possesses eigenvalues

$$\lambda_1, \lambda_2, \dots, \lambda_n$$

with corresponding eigenvectors

$$\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n.$$

If these eigenvectors are linearly independent, then

$$\delta\mathbf{X}(0) = \sum_{i=1}^n c_i \mathbf{e}_i.$$

Consequently,

$$\delta\mathbf{X}(t) = \sum_{i=1}^n c_i e^{\lambda_i t} \mathbf{e}_i.$$

Thus, the evolution of each perturbation mode is governed by its corresponding eigenvalue.

Stability Criterion The stability of an equilibrium point depends entirely on the real parts of the eigenvalues.

- If

$$\operatorname{Re}(\lambda_i) < 0, \quad \forall i,$$

all perturbations decay exponentially and the equilibrium is **asymptotically stable**.

- If at least one eigenvalue satisfies

$$\operatorname{Re}(\lambda_i) > 0,$$

the corresponding perturbation grows exponentially and the equilibrium is **unstable**.

- If one or more eigenvalues have zero real part, linearization alone is insufficient and higher-order nonlinear terms must be examined.

Hence, the stability of an equilibrium can often be determined simply from the signs of the real parts of the eigenvalues.

Routh-Hurwitz Criterion For higher-dimensional systems, finding eigenvalues analytically may be difficult because the characteristic polynomial can have a high degree. Suppose the characteristic polynomial of the Jacobian matrix is

$$\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n = 0.$$

The Routh-Hurwitz criterion provides algebraic conditions that ensure all roots of the characteristic polynomial have negative real parts. Therefore,

- it is not always necessary to compute eigenvalues explicitly,
- stability can be determined directly from the coefficients of the characteristic polynomial.

2.12 Lorenz Equations

The Lorenz system is one of the most celebrated nonlinear dynamical systems. Originally proposed by Edward Lorenz (1963) as a simplified model for atmospheric convection, it later became the first mathematical model shown to exhibit deterministic chaos. The governing equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz,\end{aligned}$$

where $\sigma > 0$ is the Prandtl number, $r > 0$ is the Rayleigh parameter, and $b > 0$ is a geometric constant. At an equilibrium point,

$$\dot{x} = \dot{y} = \dot{z} = 0.$$

Hence,

$$\begin{aligned}\sigma(y - x) &= 0, \\ rx - y - xz &= 0, \\ xy - bz &= 0.\end{aligned}$$

From the first equation,

$$y = x.$$

Substituting this into the second equation,

$$rx - x - xz = 0,$$

or

$$x(r - 1 - z) = 0.$$

Thus two possibilities arise.

Case I: $x = 0$ Thus

$$y = 0.$$

Therefore,

$$C_1 = (0, 0, 0).$$

Case II: $x \neq 0$ Then

$$z = r - 1.$$

Substituting into the third equation,

$$x = \pm \sqrt{b(r - 1)}.$$

Since

$$y = x,$$

the remaining equilibrium points are

$$C_2 = \left(\sqrt{b(r - 1)}, \sqrt{b(r - 1)}, r - 1 \right),$$

and

$$C_3 = \left(-\sqrt{b(r - 1)}, -\sqrt{b(r - 1)}, r - 1 \right).$$

Now observe that $x^2 = b(r - 1)$. Therefore,

- if

$$r < 1,$$

then

$$b(r - 1) < 0,$$

so no additional real equilibria exist. Only

$$C_1 = (0, 0, 0)$$

exists.

- if

$$r > 1,$$

three real equilibrium points exist.

Also, the Jacobian matrix of the Lorenz system is

$$J = \begin{bmatrix} -\sigma & \sigma & 0 \\ r - z & -1 & -x \\ y & x & -b \end{bmatrix}.$$

The stability of each equilibrium point is determined from the eigenvalues of this matrix.

Case I: Stability of the $(0, 0)$ The Jacobian

$$J(C_1) = \begin{bmatrix} -\sigma & \sigma & 0 \\ r & -1 & 0 \\ 0 & 0 & -b \end{bmatrix}.$$

The characteristic equation is

$$\det(J - \lambda I) = (-b - \lambda)[(-\sigma - \lambda)(-1 - \lambda) - \sigma r] = 0.$$

Therefore,

$$\lambda_1 = -b.$$

Hence,

$$\lambda_{2,3} = \frac{-(\sigma + 1) \pm \sqrt{(\sigma + 1)^2 + 4\sigma(r - 1)}}{2}.$$

When $r < 1$ The constant term

$$\sigma(1 - r) > 0.$$

Both eigenvalues have negative real parts. Since

$$\lambda_1 = -b < 0,$$

all three eigenvalues are negative. Therefore,

C_1 is asymptotically stable.

When $r > 1$ The constant term becomes negative. Hence one eigenvalue is positive. Therefore, C_1 is unstable. Suppose the characteristic polynomial is

$$\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0.$$

For a cubic equation, the equilibrium is asymptotically stable if

$$a_1 > 0, \quad a_2 > 0, \quad a_3 > 0, \quad a_1a_2 > a_3.$$

At the origin,

$$\begin{aligned} a_1 &= \sigma + b + 1, \\ a_2 &= \sigma(1 + b - r) + b, \\ a_3 &= \sigma b(1 - r). \end{aligned}$$

When

$$r < 1,$$

all four inequalities are satisfied. Hence the origin is stable. When

$$r > 1, \text{ and } a_3 < 0,$$

which immediately violates the Routh-Hurwitz criterion. Therefore,

$$C_1$$

becomes unstable.

Case II: Stability of C_2 and C_3 For

$$C_2, C_3,$$

the characteristic polynomial is again cubic. Its coefficients are

$$a_1 = \sigma + b + 1,$$

$$a_2 = b(\sigma + r),$$

$$a_3 = 2\sigma b(r - 1).$$

Since

$$r > 1,$$

the first three Routh-Hurwitz conditions are automatically satisfied. The remaining condition,

$$a_1 a_2 > a_3,$$

after simplification yields

$$r < r_c = \frac{\sigma(\sigma + b + 3)}{\sigma - b - 1}.$$

Thus,

$$1 < r < r_c$$

is the stability region of the two nonzero equilibrium points.

Classical Lorenz Parameters Lorenz used

$$\sigma = 10, \quad b = \frac{8}{3}.$$

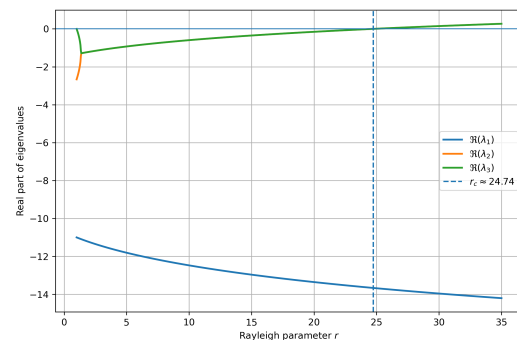
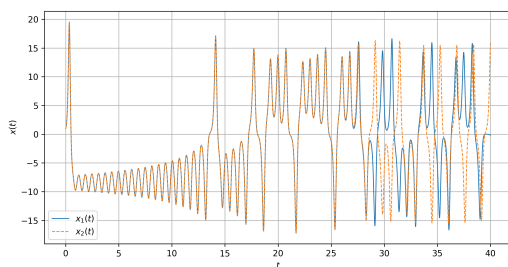
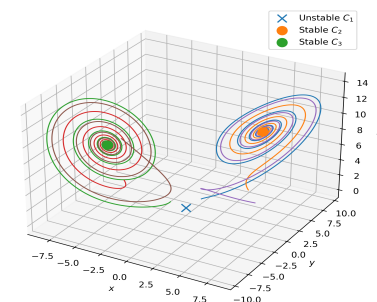
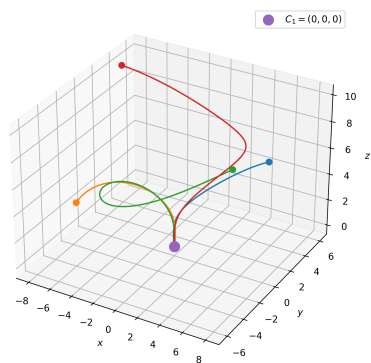
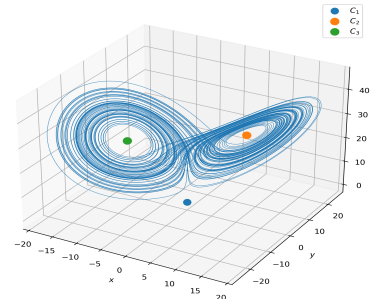
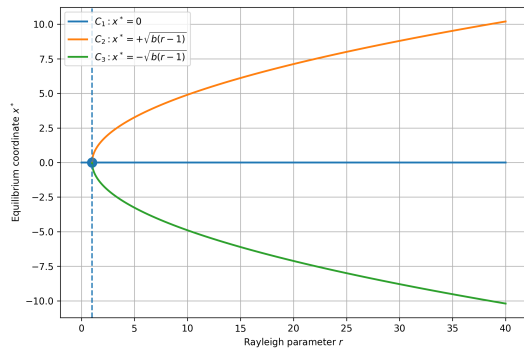
Hence

$$r_c = \frac{470}{19} \approx 24.74.$$

Therefore,

$$1 < r < 24.74 \implies C_2 \text{ and } C_3 \text{ are asymptotically stable.}$$

For $r > 24.74$ The Routh-Hurwitz criterion fails. One real eigenvalue remains negative, while the remaining two become complex conjugates with positive real parts. Hence C_2 and C_3 become unstable.



Graphical Interpretation of the Lorenz Dynamics For the classical parameters

$$\sigma = 10, \quad b = \frac{8}{3}, \quad r_c = \frac{470}{19} \approx 24.74,$$

the phase-space behavior changes significantly with the Rayleigh parameter r . For $r < 1$, the origin $C_1 = (0, 0, 0)$ is the only equilibrium and all nearby trajectories approach it. For $1 < r < r_c$, the origin becomes unstable, while the two nonzero equilibria C_2 and C_3 are asymptotically stable; trajectories approach one of these two attractors depending on the initial condition. The corresponding regions of initial conditions are called *basins of attraction*. At $r = r_c$, a complex conjugate pair of eigenvalues of C_2 and C_3 crosses the

imaginary axis, $\Re(\lambda) = 0$, and the nonzero equilibria lose stability. For $r = 28 > r_c$, all three equilibria are unstable, but trajectories may remain bounded and exhibit complicated recurrent motion around C_2 and C_3 . The resulting structure is the famous **Lorenz attractor** and illustrates **deterministic chaos**. A characteristic feature of chaotic motion is **sensitive dependence on initial conditions**. For example, trajectories starting from

$$X_1(0) = (1, 1, 1), \quad X_2(0) = (1.000001, 1, 1)$$

are initially almost indistinguishable, since

$$\|X_1(0) - X_2(0)\| = 10^{-6},$$

but eventually separate significantly. This phenomenon is commonly referred to as the *butterfly effect*. Thus, deterministic system $\nRightarrow$ predictable long-term motion.

2.13 Quasiperiodic Attractors

In nonlinear dynamical systems, the long-term motion is not always confined to an equilibrium point or a periodic orbit. Another important type of motion is called **quasiperiodic motion**. Such motions commonly arise in mechanics, electrical circuits, celestial mechanics, and nonlinear oscillators.

Quasiperiodic Motion A motion is said to be **quasiperiodic** if it is composed of two or more independent oscillations whose frequencies are incommensurate (i.e., their ratios are irrational). Mathematically, a quasiperiodic solution can be expressed as

$$x(t) = \Phi(\omega_1 t, \omega_2 t, \dots, \omega_n t),$$

where

- Φ is periodic in each of its arguments,
- $\omega_1, \omega_2, \dots, \omega_n$ are independent frequencies.

Conditions on the Frequencies For quasiperiodic motion, the frequencies must satisfy the following conditions.

1. Linear Independence There should not exist integers $k_1, k_2, \dots, k_n$, not all zero, such that $k_1\omega_1 + k_2\omega_2 + \dots + k_n\omega_n = 0$. Hence, the frequencies are linearly independent over the integers.

2. Irrational Frequency Ratios The ratios between the frequencies must be irrational. For two frequencies, $\frac{\omega_1}{\omega_2} \notin \mathbb{Q}$. Consequently, the motion never repeats exactly, and the trajectory never closes on itself.

Geometric Interpretation Each independent frequency contributes one angular coordinate. Therefore,

- one frequency corresponds to motion on a circle,
- two independent frequencies correspond to motion on a two-dimensional torus,
- three independent frequencies correspond to motion on a three-dimensional torus.

In general, n independent frequencies $\implies n$ -dimensional torus (T^n).

Quasiperiodic Attractor If nearby trajectories approach a torus carrying quasiperiodic motion, then the torus acts as an attractor. Stable Torus $\implies$ Quasiperiodic Attractor. Unlike a limit cycle, the motion never repeats exactly, infinitely many oscillation phases coexist, and trajectories densely cover the attracting torus.

Periodic or Quasiperiodic Motion Consider the motion

$$x(t) = b_1 \cos(\omega_1 t) + b_2 \cos(\omega_2 t),$$

where b_1 and b_2 are constants. Two cases arise.

Case 1: Rational Frequency Ratio If

$$\frac{\omega_1}{\omega_2} = \frac{m}{n},$$

where m and n are integers, then both oscillations repeat after a common period. Hence, motion is periodic. The corresponding phase trajectory is a closed curve.

Case 2: Irrational Frequency Ratio If

$$\frac{\omega_1}{\omega_2}$$

is irrational, the oscillations never repeat simultaneously. Therefore, the motion is quasiperiodic. The phase trajectory never closes and gradually fills the invariant torus.

Example Consider

$$x(t) = \cos t + \cos(\sqrt{2}t).$$

Here,

$$\omega_1 = 1, \quad \omega_2 = \sqrt{2}.$$

Since

$$\frac{\omega_1}{\omega_2} = \frac{1}{\sqrt{2}}$$

is irrational, the motion is quasiperiodic. The corresponding phase portrait is not a closed orbit.

Example: Quasiperiodic Motion on an Attracting Torus Consider the dynamical system in toroidal coordinates

$$\begin{aligned}\dot{\rho} &= 1 - \rho, \\ \dot{\theta} &= 1, \\ \dot{\phi} &= \sqrt{2},\end{aligned}$$

where ρ measures the distance from an invariant torus and θ, ϕ are angular variables. The first equation has the solution

$$\rho(t) = 1 + (\rho_0 - 1)e^{-t}.$$

Hence,

$$\lim_{t \rightarrow \infty} \rho(t) = 1.$$

Therefore, all nearby trajectories approach the surface $\rho = 1$. Thus, the torus $\rho = 1$ is an attracting invariant set. On the torus,

$$\theta(t) = \theta_0 + t, \quad \phi(t) = \phi_0 + \sqrt{2}t.$$

The two angular frequencies are

$$\omega_1 = 1, \quad \omega_2 = \sqrt{2}.$$

Since

$$\frac{\omega_1}{\omega_2} = \frac{1}{\sqrt{2}} \notin \mathbb{Q},$$

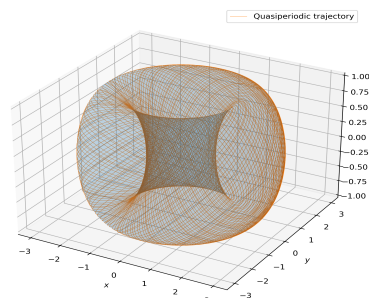
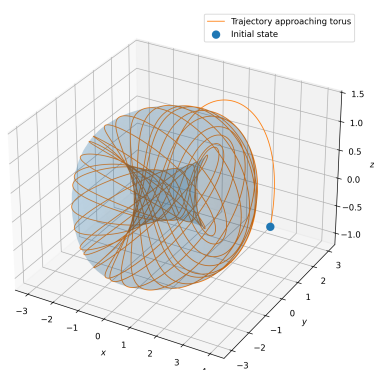
the two angular motions never repeat simultaneously. Consequently, the trajectory never closes on itself. Instead, as $t \rightarrow \infty$, the trajectory moves quasiperiodically and densely covers the invariant torus. Hence, Attraction to a torus + incommensurate frequencies $\Rightarrow$ quasiperiodic attractor.

To visualize the motion, a torus may be represented in Cartesian coordinates by

$$\begin{aligned} x &= (R + \rho \cos \phi) \cos \theta, \\ y &= (R + \rho \cos \phi) \sin \theta, \\ z &= \rho \sin \phi, \end{aligned}$$

where $R > 1$ is the major radius of the torus. A trajectory starting with $\rho_0 \neq 1$ first approaches the torus because $\rho(t) \rightarrow 1$. After the transient motion disappears, the trajectory continues to move with the two independent frequencies 1 and $\sqrt{2}$. Unlike a stable equilibrium, the trajectory does not approach a point. Unlike a stable limit cycle, the motion does not approach a single closed orbit. Instead, the attracting set is an entire torus carrying quasiperiodic motion. Thus,

Stable equilibrium $\rightarrow$ point attractor,
Stable limit cycle $\rightarrow$ periodic attractor,
Stable invariant torus $\rightarrow$ quasiperiodic attractor.



2.14 Chaotic Attractors

The attractors studied so far, namely equilibrium points, limit cycles and quasiperiodic attractors, describe motions that are stationary, periodic or almost periodic. In contrast, many nonlinear dynamical systems exhibit a more complicated type of long-term behavior known as **chaotic motion**.

Definition A **chaotic attractor** is a bounded attracting set on which the system evolves in an apparently irregular and unpredictable manner. Although the motion is deterministic (governed entirely by differential equations), it appears random because of its extreme sensitivity to initial conditions.

Unlike equilibrium points or periodic orbits, a chaotic attractor is neither a single point, a closed orbit, nor an invariant torus.

Characteristics of Chaotic Attractors A chaotic attractor possesses the following important properties.

1. **Bounded Motion** The trajectory always remains inside a finite region of the phase space.
2. **Aperiodic Motion** The trajectory never repeats itself exactly and does not possess a finite period.
3. **Sensitive Dependence on Initial Conditions** Two trajectories starting from very close initial conditions separate exponentially with time.
4. **Deterministic Dynamics** Although the motion appears random, it is completely governed by the underlying differential equations.

Sensitive Dependence on Initial Conditions The defining feature of chaos is the sensitive dependence on initial conditions. Suppose two nearby trajectories have initial conditions

$$\mathbf{x}_1(0), \quad \mathbf{x}_2(0),$$

with initial separation

$$\delta(0) = \|\mathbf{x}_1(0) - \mathbf{x}_2(0)\|.$$

For a chaotic system,

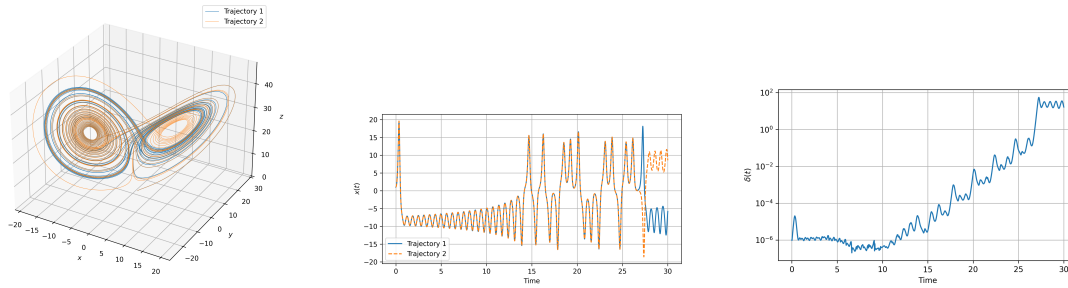
$$\delta(t) \approx \delta(0)e^{\lambda t},$$

where $\lambda > 0$ is called the **largest Lyapunov exponent**. Since $e^{\lambda t}$ grows rapidly, even extremely small differences in the initial conditions eventually produce completely different trajectories. Although the governing equations are deterministic, measurements of the initial conditions always contain small errors. If $\delta(0)$ denotes the measurement error, then after time t ,

$$\delta(t) = \delta(0)e^{\lambda t}.$$

Hence the prediction error grows exponentially. Consequently, long-term prediction becomes practically impossible even though the system itself is deterministic.

Lorentz Oscillator See the explanation for the Lorentz oscillator previously after the section on higher dimensional dynamical systems.



2.15 Poincaré Map

The continuous evolution of a nonlinear dynamical system is often difficult to analyze, especially when the trajectories become complicated. A useful technique for simplifying the study of such systems is the **Poincaré map**, which converts a continuous-time dynamical system into a discrete-time system.

Definition A **Poincaré map** is obtained by observing the state of a dynamical system at fixed intervals of time or whenever the trajectory intersects a chosen surface (called the Poincaré section). Instead of studying a continuous trajectory, we analyze the sequence of points

$$X_0, X_1, X_2, \dots$$

generated by successive observations. Thus, a continuous flow is transformed into a discrete map.

Why Use a Poincaré Map? The Poincaré map reduces the dimensionality of the problem and makes the long-term behavior of the system easier to visualize. It is particularly useful for identifying periodic motion, detecting quasiperiodic motion, distinguishing chaotic attractors, and studying bifurcations.

Poincaré Map of a Quasiperiodic Motion Consider the quasiperiodic solution

$$x(t) = b_1 \cos(\omega_1 t) + b_2 \cos(\omega_2 t),$$

where $\frac{\omega_1}{\omega_2}$ is irrational. To construct the Poincaré map, sample the solution at the times

$$t_n = \frac{2\pi n}{\omega_1}, \quad n = 0, 1, 2, \dots$$

Since $\cos(\omega_1 t_n) = 1$, the sampled coordinates become

$$x_n = b_1 + b_2 \cos\left(\frac{2\pi n \omega_2}{\omega_1}\right),$$

and

$$y_n = -b_2 \omega_2 \sin\left(\frac{2\pi n \omega_2}{\omega_1}\right).$$

The collection of points (x_n, y_n) forms the Poincaré map. Now to analyze the shape of the Poincaré map, we square the above equations,

$$(x_n - b_1)^2 = b_2^2 \cos^2 \theta,$$

where $\theta = \frac{2\pi n\omega_2}{\omega_1}$, and

$$\left(\frac{y_n}{\omega_2}\right)^2 = b_2^2 \sin^2 \theta.$$

Adding the two equations,

$$(x_n - b_1)^2 + \left(\frac{y_n}{\omega_2}\right)^2 = b_2^2.$$

This is the equation of an ellipse. Hence, the Poincaré map of a quasiperiodic motion is a closed ellipse (or, in general, a closed invariant curve).

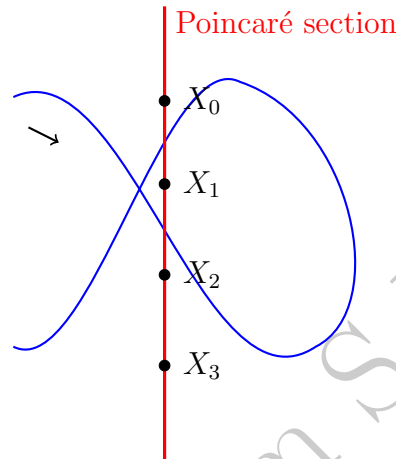
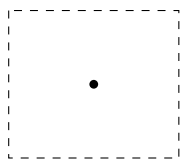


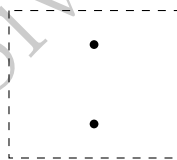
Figure 7: A continuous trajectory intersects the Poincaré section repeatedly. Only the intersection points are retained to construct the Poincaré map.

(a) Periodic



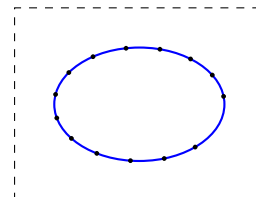
One point

(b) Period-2



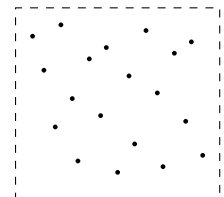
Two points

(c) Quasiperiodic



Invariant curve

(d) Chaotic



Irregular set

Figure 8: Typical Poincaré maps. (a) A periodic orbit produces a single point. (b) A period-two orbit produces two alternating points. (c) A quasiperiodic orbit generates an invariant closed curve. (d) A chaotic orbit produces an irregular (fractal) set of points.

2.16 Dissipative and Conservative Systems

In dynamical systems, an important classification is based on whether the system conserves energy or dissipates it during motion. This leads to the concepts of **conservative** and **dissipative** systems.

Conservative Systems A dynamical system is called **conservative** if the total mechanical energy remains constant during the motion. Suppose a particle is subjected to a force

$$\mathbf{F} = -\nabla V,$$

where $V(x, y, z)$ is the potential energy, V depends only on position, and ∇ denotes the gradient operator. Such a force is called a **conservative force**.

Work Done by a Conservative Force Let the particle undergo an infinitesimal displacement $d\mathbf{r}$. The elementary work done is $dW = \mathbf{F} \cdot d\mathbf{r}$. Using

$$\mathbf{F} = -\nabla V,$$

we obtain

$$dW = -\nabla V \cdot d\mathbf{r}.$$

Since

$$dV = \nabla V \cdot d\mathbf{r},$$

it follows that

$$dW = -dV.$$

Integrating between two points A and B ,

$$W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = -\int_A^B dV.$$

Therefore,

$$W = V(A) - V(B).$$

Hence, the work done depends only on the initial and final points, and it is independent of the path followed. If the particle returns to its starting point, $A = B$, then $W = V(A) - V(A) = 0$. Therefore,

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0.$$

This proves that the work done around any closed path is zero. Hence the total mechanical energy $E = T + V$ remains constant. Such systems are called conservative systems.

Phase Space Volume Consider an n -dimensional dynamical system

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}).$$

Suppose a small rectangular volume around a point in phase space is

$$\Delta T = \prod_{i=1}^n \Delta x_i.$$

As time evolves, this volume may remain constant, or shrink, or expand. Studying this behavior helps classify dynamical systems.

Rate of Change of Phase Volume Taking logarithms,

$$\ln(\Delta T) = \sum_{i=1}^n \ln(\Delta x_i).$$

Differentiating with respect to time,

$$\frac{1}{\Delta T} \frac{d(\Delta T)}{dt} = \sum_{i=1}^n \frac{1}{\Delta x_i} \frac{d(\Delta x_i)}{dt}.$$

For small perturbations,

$$\frac{d(\Delta x_i)}{dt} = \sum_{j=1}^n \frac{\partial F_i}{\partial x_j} \Delta x_j.$$

Evaluating the instantaneous change at the chosen point,

$$\frac{1}{\Delta T} \frac{d(\Delta T)}{dt} = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i}.$$

The quantity

$$\nabla \cdot \mathbf{F} = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i}$$

is called the **divergence** of the vector field.

Classification Using Divergence The divergence completely determines the local behavior of phase-space volume.

Case 1 If

$$\nabla \cdot \mathbf{F} = 0,$$

then

$$\frac{d(\Delta T)}{dt} = 0.$$

Hence,

$$\Delta T = \text{constant}.$$

The phase-space volume is preserved. The system is called Conservative.

Case 2 If

$$\nabla \cdot \mathbf{F} < 0,$$

then

$$\Delta T$$

decreases continuously. Nearby trajectories move closer together. The system loses energy. Hence, a dissipative system

Case 3 If

$$\nabla \cdot \mathbf{F} > 0,$$

phase-space volume increases. Nearby trajectories move apart. Such systems are called Volume Expanding Systems

Example Consider

$$\ddot{x} + f(x) = 0.$$

Introduce

$$x_1 = x, \quad x_2 = \dot{x}.$$

Then

$$\dot{x}_1 = x_2, \text{ and } \dot{x}_2 = -f(x_1).$$

Hence

$$F_1 = x_2, \quad F_2 = -f(x_1).$$

Now compute the divergence.

$$\frac{\partial F_1}{\partial x_1} = 0,$$

because x_2 is independent of x_1 . Also,

$$\frac{\partial F_2}{\partial x_2} = 0,$$

because F_2 depends only on x_1 . Therefore,

$$\nabla \cdot F = 0 + 0 = 0.$$

Hence, the system is conservative.

Example Consider

$$\ddot{x} + d\dot{x} + f(x) = 0, \quad d > 0.$$

Let

$$x_1 = x, \quad x_2 = \dot{x}.$$

Then

$$\dot{x}_1 = x_2, \text{ and } \dot{x}_2 = -dx_2 - f(x_1).$$

Hence,

$$F_1 = x_2, \text{ and } F_2 = -dx_2 - f(x_1).$$

Now,

$$\frac{\partial F_1}{\partial x_1} = 0,$$

and

$$\frac{\partial F_2}{\partial x_2} = -d.$$

Therefore,

$$\nabla \cdot F = -d < 0.$$

Hence, the system is dissipative. Since $d > 0$, the phase-space volume decreases exponentially.

Example Show that the Lorenz system is dissipative. The Lorenz equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz.\end{aligned}$$

The vector field is

$$\begin{aligned}F_1 &= \sigma(y - x), \\ F_2 &= rx - y - xz, \\ F_3 &= xy - bz.\end{aligned}$$

Compute each partial derivative.

$$\frac{\partial F_1}{\partial x} = -\sigma, \quad \frac{\partial F_2}{\partial y} = -1, \quad \text{and} \quad \frac{\partial F_3}{\partial z} = -b.$$

Therefore,

$$\nabla \cdot F = -\sigma - 1 - b.$$

Since

$$\sigma > 0, \quad b > 0,$$

we obtain

$$\nabla \cdot F < 0.$$

Hence, the Lorenz system is dissipative. The phase-space volume contracts exponentially with time.

Remark Many nonlinear systems possessing stable equilibrium points, stable limit cycles, or chaotic attractors are dissipative. The contraction of phase-space volume allows trajectories starting from different initial conditions to approach the same attracting set.

In contrast, conservative systems preserve phase-space volume. Consequently, they cannot possess attracting or repelling limit cycles. Instead, they usually contain continuous families of periodic orbits, each corresponding to a different initial condition.

That is, conservative systems generally exhibit regular periodic motion, whereas dissipative systems may possess stable equilibrium points, limit cycles, quasiperiodic attractors, and even chaotic attractors.

Comparison of Conservative and Dissipative Systems **Conservative System.** Consider the simple harmonic oscillator

$$\ddot{x} + x = 0. \quad \text{the system becomes} \quad \dot{x}_1 = x_2, \quad \dot{x}_2 = -x_1.$$

Thus, the divergence is

$$\nabla \cdot F = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} = 0 + 0 = 0.$$

Hence, the phase-space volume is preserved, and the system is conservative. The phase portrait consists of a family of closed curves (circles or ellipses), each corresponding to a different initial condition.

Dissipative System. Consider the damped harmonic oscillator

$$\ddot{x} + c\dot{x} + x = 0, \quad c > 0. \quad \text{Thus} \quad \dot{x}_1 = x_2, \quad \dot{x}_2 = -x_1 - cx_2.$$

Hence, the divergence is

$$\nabla \cdot F = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} = 0 - c = -c < 0.$$

Therefore, the phase-space volume contracts with time, and the system is dissipative. All trajectories spiral toward the stable equilibrium at the origin because energy is continuously lost due to damping.

Property	Conservative System	Dissipative System
Mechanical Energy	Constant	Decreases with time
Divergence	$\nabla \cdot F = 0$	$\nabla \cdot F < 0$
Phase-Space Volume	Preserved	Contracts
Typical Trajectories	Families of closed orbits	Spirals toward attractors
Examples	Simple harmonic oscillator	Damped oscillator, Lorenz system
Stable Limit Cycle	Not possible	Possible
Chaotic Attractor	Not possible	Possible

2.17 Hamiltonian Systems

Hamiltonian systems form one of the most important classes of conservative dynamical systems. They arise naturally in classical mechanics and describe the evolution of systems in which the total energy is conserved. Unlike dissipative systems, Hamiltonian systems preserve phase-space volume during their evolution.

Generalized Coordinates and Momenta Consider a mechanical system with n degrees of freedom. Instead of using only the generalized coordinates

$$q_1, q_2, \dots, q_n,$$

Hamiltonian mechanics introduces the corresponding generalized momenta

$$p_1, p_2, \dots, p_n.$$

Thus, the complete state vector becomes

$$X = (q_1, q_2, \dots, q_n, p_1, p_2, \dots, p_n)^T,$$

which belongs to a $2n$ -dimensional phase space.

Hamilton's Equations The dynamics of a Hamiltonian system are governed by Hamilton's equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i},$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial q_i},$$

where

$$i = 1, 2, \dots, n,$$

and

$$H(q, p)$$

is called the Hamiltonian function. In many mechanical systems,

$$H = T + V,$$

where

- T is the kinetic energy,
- V is the potential energy.

Phase-Space Volume Conservation One of the most remarkable properties of Hamiltonian systems is that the phase-space volume remains constant during the motion. To prove this, consider the vector field

$$F = (\dot{q}_1, \dots, \dot{q}_n, \dot{p}_1, \dots, \dot{p}_n).$$

The divergence of the vector field is

$$\nabla \cdot F = \sum_{i=1}^n \frac{\partial \dot{q}_i}{\partial q_i} + \sum_{i=1}^n \frac{\partial \dot{p}_i}{\partial p_i}.$$

Substituting Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i},$$

and

$$\dot{p}_i = -\frac{\partial H}{\partial q_i},$$

gives

$$\nabla \cdot F = \sum_{i=1}^n \frac{\partial}{\partial q_i} \left(\frac{\partial H}{\partial p_i} \right) + \sum_{i=1}^n \frac{\partial}{\partial p_i} \left(-\frac{\partial H}{\partial q_i} \right).$$

Hence,

$$\nabla \cdot F = \sum_{i=1}^n \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right).$$

If the Hamiltonian possesses continuous second-order partial derivatives, Clairaut's theorem gives

$$\frac{\partial^2 H}{\partial q_i \partial p_i} = \frac{\partial^2 H}{\partial p_i \partial q_i}.$$

Therefore,

$$\nabla \cdot F = 0.$$

Hence,

$$\frac{d(\Delta V)}{dt} = 0,$$

which means that every infinitesimal phase-space volume remains unchanged. This result is known as Liouville's Theorem.

Physical Interpretation Suppose a closed surface encloses a small volume in phase space. As every point on the surface evolves according to Hamilton's equations, the shape of the volume may deform, the volume may stretch or bend, but its total volume always remains constant. Hence Hamiltonian systems preserve phase-space volume. The types of equilibrium points also differ. Hamiltonian systems can possess Centers, or Saddles. They cannot possess Stable nodes, Stable stars, Stable spirals (foci), Unstable nodes, Unstable stars, or Unstable spirals. Since

$$\nabla \cdot F = 0,$$

the sum of the eigenvalues of the Jacobian matrix is zero. Therefore,

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = 0.$$

Consequently, all eigenvalues cannot simultaneously have negative real parts. Hence asymptotically stable equilibria cannot exist. Similarly, all eigenvalues cannot simultaneously have positive real parts. Thus, only centers and saddles are possible equilibrium points.